

Leopold-Franzens University of Innsbruck
Faculty of Mathematics, Computer Science and Physics

Institute for Astro- and Particle Physics

Extragalactic Astrophysics Research Group



Bachelor's thesis

to achieve the academic degree

Bachelor of Science

Accelerating Models of the Connection of Dark Matter and Galaxies with Deep Neural Networks

by

Robin Sebastian Hoffmann
(Matr.-Nr.: 12104380)

Submission Date: July 15, 2025
Supervisor: Dr. Laila Linke

Declaration of Honour

I hereby declare in lieu of oath with my personal signature that I have written this thesis independently and have not used any sources or aids other than those indicated. All passages that have been taken verbatim or in terms of content from the stated sources are marked as such. I agree to the archiving of this bachelor thesis.

Place and Date

Stuttgart, 24.08.2025

Signature:

R. Hoffmann

Contents

List of Figures	II
Abstract	1
1 Introduction	2
2 Theoretical Background	4
2.1 Physical Models	4
2.1.1 Cold Dark Matter	4
2.1.2 Λ CDM Model	5
2.1.3 Density Contrast	7
2.1.4 Halo Model	8
2.1.5 Linear Matter Power Spectrum	13
2.1.6 Non-linear Power Spectrum	13
2.1.7 Halo Occupation Distribution	15
2.2 Neural Networks	18
2.2.1 Surrogate Models	18
2.2.2 Architecture	19
2.2.3 Learning	21
2.2.4 Training	23
2.2.5 Hyperparameter Optimization	25
2.2.6 Predictive Modeling for Computational Efficiency	26
3 Realization and Implementation	28
3.1 Data Generation	28
3.2 Data Preprocessing	29
3.3 Neural Network Training	31
3.4 Neural Network Finetuning	33
4 Results	40
4.1 Accuracy of Predictions	40
4.2 Performance of Predictions	44
4.3 Code	46
5 Discussion	47
References	50

List of Figures

1	Halo Mass Function	9
2	Halo Bias	11
3	Halo Profile	12
4	Non-linear Power Spectrum	14
5	Halo Occupation Distribution	17
6	NN Architecture	20
7	Activation Functions	21
8	Training Dataset Distribution	35
9	Cross Check	36
10	Training Process	37
11	Optimizer Hyperparameter Evaluation	38
12	Parameter Space with Sample Error	39
13	Prediction Accuracy	42
14	Relative Error of Predictions	42
15	Model Comparison Relative Error 1- and 2-halo Term	43
16	Model Comparison Relative Error of Total Term	43
17	CPU-Time Comparison	45
18	CPU-Time Model Comparison	45

Abstract

In this thesis, neural networks are trained to predict two-point statistics of the large-scale structure, including matter–matter, galaxy–matter, and galaxy–galaxy power spectra, based on the halo model. This model is combined with a halo occupation distribution that captures the connection between dark matter and galaxies, providing orders-of-magnitude acceleration for cosmological parameter estimation. The predicted spectra exhibit errors below 1.1% for 99% of the spectra in the wavenumber range $k \in [10^{-2}, 10] \text{ Mpc}^{-1}$ at redshift $z = 0$. The neural networks provide a computational speed-up of approximately $O(10^4)$ compared to traditional model evaluations. This acceleration allows rapid reconstruction of posterior distributions, enabling efficient estimation of cosmological parameters.

1 Introduction

Understanding the origin, structure, and evolution of our Universe has long captivated the curiosity of humankind. Modern cosmology has developed remarkably successful models — most notably the Λ CDM paradigm — which provide a coherent framework for describing cosmic history and the large-scale structure of the Universe. These models are not static; they are continuously refined and extended as new data and theoretical insights become available. Yet, fundamental questions remain, such as the nature of dark matter, the connection between the dark matter distribution and galaxy formation, and persistent tensions in observational data, exemplified by the so-called Hubble tension (Del Popolo & Le Delliou 2017). Addressing these open problems requires not only increasingly precise measurements but also more accurate modeling of the Universe in order to robustly constrain cosmological parameters. In this context, the era of artificial intelligence opens up new opportunities: computationally intensive modeling can be significantly accelerated through machine-learning-based predictions, enabling faster and more flexible exploration of cosmological parameter spaces (Piras et al. 2024, Spurio Mancini et al. 2022).

This work aims to accelerate the parameter inference of cosmological models with Bayesian posterior sampling by speeding up the sampling process using neural networks. The key observable in this context is the power spectrum, which quantifies the clustering of matter and galaxies across different spatial scales and serves as a crucial link between theoretical models and observational data. Comparing power

1 Introduction

spectra for various cosmological parameter configurations with observations enables the inference of the underlying model parameters. However, computing these power spectra with high accuracy — especially when using advanced approaches such as the halo model combined with a halo occupation distribution to describe the connection between dark matter and galaxies — is computationally expensive. To overcome this limitation, we train neural networks to emulate predictions of the power spectrum for different cosmological configurations. Once trained, these networks can deliver accurate results orders of magnitude faster than conventional methods, making large-scale sampling practical. A central goal of this work is therefore to identify ideal neural network architectures that can accurately and computationally efficiently predict power spectra.

This thesis first introduces the physical models employed in Section 2.1, with a focus on the Λ CDM framework and the halo model, including the halo occupation distribution. It then describes the functionality and advantages of neural networks in this context, as detailed in Section 2.2, together with the training strategies used, which are presented in Chapter 3. Finally, it analyzes the results in Chapter 4, demonstrating the performance of the trained neural networks and evaluating their applicability for accelerating cosmological parameter inference, which is further discussed in Chapter 5.

2 Theoretical Background

2.1 Physical Models

The content of this work is fundamentally based on the physical models presented in this chapter, which provide the theoretical framework for all subsequent modeling and interpretation. These models include the Cold Dark Matter (CDM) model, the Λ CDM model, the halo model, and the Halo Occupation Distribution (HOD).

2.1.1 Cold Dark Matter

Dark matter refers to non-baryonic matter that does not interact electromagnetically and is therefore only observable through its gravitational effects. This additional component, distinct from the baryonic matter that is directly detectable via electromagnetic radiation, is required to explain several key astrophysical and cosmological observations.

One of the earliest and most compelling pieces of evidence for dark matter arises from the rotation curves of spiral galaxies. These rotation curves do not exhibit the expected Keplerian fall-off at large radii, which implies the presence of a substantial, invisible mass component extending beyond the luminous baryonic matter (Persic et al. 1996).

Gravitational lensing also provides strong support for dark matter. The observed bending of light from distant sources by foreground mass distributions often exceeds

2 Theoretical Background

what would be expected from visible matter alone. Additionally, fluctuations in the Cosmic Microwave Background (CMB) and the formation of large-scale structures in the Universe are consistent only when dark matter is included in the cosmological model. Without a non-relativistic, gravitationally interacting component, the growth of initial density perturbations in the early Universe would be insufficient to yield the cosmic web of galaxies and clusters observed today (Vegetti et al. 2023).

The Cold Dark Matter (CDM) model describes dark matter as non-relativistic during the relevant epochs of structure formation. Its exact nature and composition remain unknown — including whether it consists of particles, often associated with hypothetical weakly interacting massive particles (WIMPs), or more exotic objects like primordial black holes.

A key prediction we take of the CDM model is the hierarchical, bottom-up structure formation. In this scenario, low-mass halos form first and gradually merge to create increasingly massive and complex structures. The collisionless nature of CDM combined with this hierarchical formation leads to a population of dark matter halos with well-defined statistical properties, which manifest observationally through rotation curves, gravitational lensing, and anisotropies in the CMB (Vegetti et al. 2023).

2.1.2 Λ CDM Model

The Λ CDM model represents the current standard model of cosmology, describing the evolution of the Universe from the Big Bang to its present state using only six cosmological parameters within the cold dark matter framework. It has proven to be a highly effective model providing a remarkably accurate fit to a wide range of cosmological observations over many decades, spanning spatial scales from megaparsecs up to the Hubble scale, the distance over which cosmic expansion dominates, and temporal scales across more than 10 Gyr of cosmic time (Planck Collaboration 2020).

2 Theoretical Background

It explains the accelerated expansion of the Universe by introducing a cosmological constant Λ , which is interpreted as dark energy in the framework of General Relativity. This can be understood as a positive vacuum energy density responsible for driving the acceleration (Bamba et al. 2012).

The expansion is characterized by the scale factor $a(t)$, which describes the relative growth of distances between comoving objects in the Universe, i.e., objects that move with the general cosmic expansion and remain at fixed positions in comoving coordinates. It is defined as

$$a(t) = \frac{R(t)}{R(t_0)}, \quad (1)$$

where $R(t)$ denotes the physical distance between two comoving galaxies at time t and $R(t_0)$ is their distance at the present time t_0 .

The scale factor is directly related to the observed redshift z , which quantifies the wavelength stretching of electromagnetic signals due to the cosmic expansion and thus explains why most observed signals from distant sources appear shifted toward longer wavelengths as these sources recede from the observer. This relation is given by

$$1 + z = \frac{1}{a(t)}. \quad (2)$$

The evolution of the scale factor can be quantified by the Hubble parameter. The Hubble parameter $H(t)$ measures the relative expansion rate of the Universe and is defined as

$$H(t) = \frac{\dot{a}(t)}{a(t)}, \quad (3)$$

where the dot denotes the time derivative.

The expansion rate of the Universe is given by the density parameter describing its composition. The Hubble parameter as a function of the scale factor can be expressed as

$$H^2(a) = H_0^2 [\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_k a^{-2} + \Omega_\Lambda], \quad (4)$$

2 Theoretical Background

where Ω_{m} , Ω_{r} , Ω_{k} , and Ω_{Λ} denote the density parameters of matter, radiation, spatial curvature, and the cosmological constant, respectively. The respective powers of a arise from the scaling behavior of the different energy components with the expansion.

A complete description of the Λ CDM model involves six key parameters: the present-day Hubble parameter H_0 ; the baryon density Ω_{b} and cold dark matter density Ω_{c} ; the amplitude A_s and spectral index n_s of the primordial curvature perturbations; and the optical depth to reionization τ . These parameters encapsulate both the background evolution and the initial conditions for structure formation in the Universe.

2.1.3 Density Contrast

The linear density contrast $\delta(\mathbf{x}, z)$ describes the relative deviation of the local matter density from the cosmic mean density $\bar{\rho}(z)$ at a given redshift z (Asgari et al. 2023). It is defined as

$$\delta(\mathbf{x}, z) = \frac{\rho(\mathbf{x}, z) - \bar{\rho}(z)}{\bar{\rho}(z)}, \quad (5)$$

where $\rho(\mathbf{x}, z)$ denotes the local matter density and $\bar{\rho}(z)$ the mean matter density of the Universe at redshift z . According to the spherical collapse model, a region is expected to collapse and form a gravitationally bound object, such as a halo, if its linearly extrapolated density contrast exceeds the critical threshold $\delta_{\text{c}}(z)$. This critical value is approximately $\delta_{\text{c}} \approx 1.69$ in an Einstein–de Sitter universe, i.e., a flat Λ CDM cosmology with matter density parameter $\Omega_{\text{m}} = 1$ (Asgari et al. 2023, Linke et al. 2022).

2.1.4 Halo Model

The halo model presents a simplified model for the large-scale structure of the Universe. It describes the complex structure of the matter distribution in our Universe, the so called „cosmic web“, as a sum of its individual components. All matter components, such as dark matter, gas, and galaxies, are assumed to be located within so-called halos. These halos are self-bound, spherical structures, defined to have an average density that is 200 times greater than the mean density of the Universe, $\bar{\rho}$. More precisely, halos form in regions at redshift z where the linear density contrast exceeds the critical density contrast $\delta_c(z)$. The properties of these halos are described by a set of simulation-calibrated functions that serve as the fundamental ingredients of the model. These include the linear matter power spectrum, the halo mass function, the linear halo bias, and the halo density profile (Asgari et al. 2023, Linke et al. 2022).

In this work, the halo model is used to model the statistics of the large-scale structure of the Universe for different cosmological configurations, with the analysis restricted to redshift zero.

Halo Mass Function

The halo mass function (HMF), denoted by $n(m, z) dm$, describes the mean comoving number density of dark matter halos with masses in the interval $[m, m + dm]$ for different redshift z as illustrated in Fig. 1. To satisfy the fundamental assumption of the halo model that all matter in the Universe is bound in halos, the HMF must fulfill the normalization condition

$$\int_0^\infty M n(M) dM = \bar{\rho}, \quad (6)$$

where $\bar{\rho}$ denotes the mean matter density of the Universe (Asgari et al. 2023).

Several forms of the HMF have been proposed in the literature, each calibrated

2 Theoretical Background

against N -body simulations. In this work, we adopt the widely used parametrization introduced by Sheth & Tormen (1999):

$$n(m, z) dm = A \frac{\bar{\rho}}{m^2} \frac{d \ln \nu}{d \ln m} dm \left[1 + \frac{1}{(q\nu)^{2p}} \right]^r \frac{q\nu}{\sqrt{2\pi}} \exp \left(-\frac{(q\nu)^2}{2} \right), \quad (7)$$

where the parameters $A = 0.322$, $p = 0.3$, and $q = 0.707$ were determined from N -body simulations. The peak-height parameter ν is defined as $\nu = \delta_c / \sigma(r_m, z)$, where δ_c is the critical density contrast for collapse and r_m is the Lagrangian radius associated with mass m , given by $r_m^3 = \frac{3m}{4\pi\bar{\rho}}$ (Linke et al. 2022).

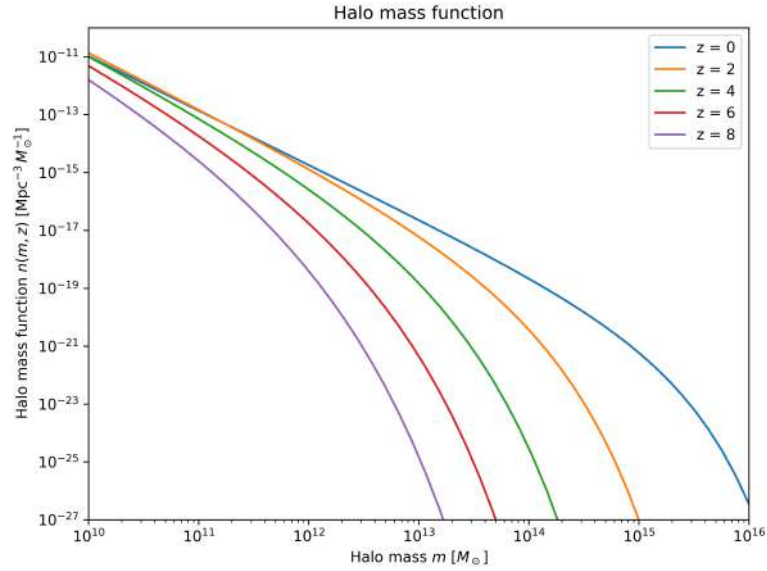


Figure 1: The Sheth-Tormen halo mass function describes the mean comoving number density of dark matter halos with masses m and redshift z (Laila Linke, personal communication, 2025).

2 Theoretical Background

Linear Halo Bias

The linear halo bias describes the spatial distribution of dark matter halos relative to the underlying matter density field in the Universe. Specifically, it quantifies the enhanced clustering of dark matter halos through the ratio of the halo density contrast δ_h to the matter density contrast δ . Both theoretical considerations and observations indicate that halos are more strongly clustered than the dark matter itself. In the halo model, this effect is described by a bias factor, defined as

$$\delta_h(\mathbf{x}, z | m) = b_1(m, z) \delta(\mathbf{x}, z), \quad (8)$$

where the bias factor $b_1(m, z)$ depends only on halo mass m and redshift z (Linke et al. 2022). Additionally, the mass-weighted bias must satisfy the normalization condition

$$\int_0^\infty M b(M) n(M) dM = \bar{\rho}, \quad (9)$$

analogous to the condition in Eq. (6) (Asgari et al. 2023).

In this work, we adopt the linear bias b_1 derived from the peak-background split formalism (Mo & White 1994, Scoccimarro et al. 2001):

$$b_1(m, z) = 1 + \frac{q\nu^2(z) - 1}{\delta_c(z)} + \frac{2p}{\delta_c(z)} \frac{1}{1 + (q\nu^2(z))^p}, \quad (10)$$

where the parameters q and p are the same as in the Sheth–Tormen HMF.

2 Theoretical Background

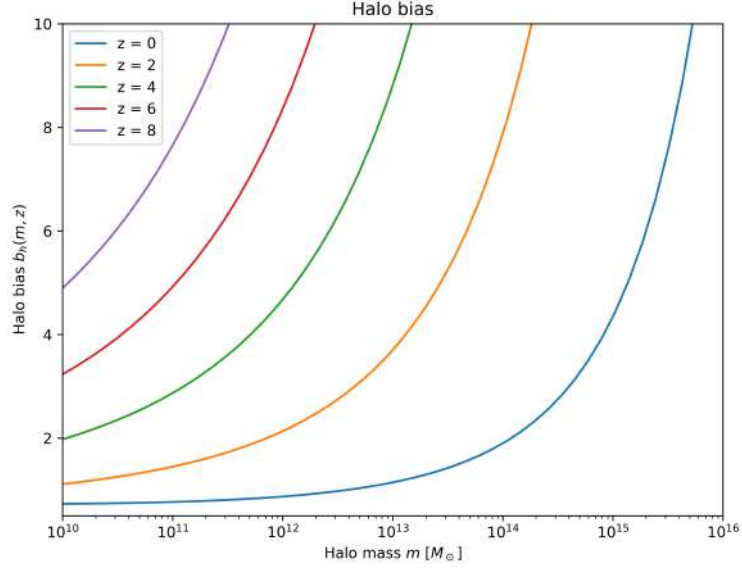


Figure 2: The linear halo bias describes the distribution of dark matter halos relative to the matter density field for halos of mass m and redshift z (Laila Linke, personal communication, 2025).

Halo Profile

The distribution of matter and galaxies within a dark matter halo can be described by universal halo density profiles. In this work, we use the Navarro–Frenk–White (NFW) profile, which depends solely on the halo mass m and the radial distance r from the halo center. The NFW profile was first introduced by Navarro et al. (1997), based on high-resolution N -body simulations, and is given by

$$\rho(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right) \left(1 + \frac{r}{r_s}\right)^2}, \quad \text{with} \quad r_s = \frac{r_{200}}{c_{200}(m, z)}. \quad (11)$$

Here, r_s denotes the scale radius, and r_{200} is the virial radius of the halo, defined such that the average density within r_{200} equals 200 times the mean matter density of the Universe at redshift z :

2 Theoretical Background

$$\bar{\rho}_{\text{halo}}(< r_{200}) = 200 \bar{\rho}. \quad (12)$$

The shape of the NFW profile is determined by the concentration parameter c_{200} , for which we adopt the mass- and redshift-dependent relation proposed by Duffy et al. (2008):

$$c_{200}(m, z) = A \left(\frac{m}{M_{\text{pivot}}} \right)^B (1+z)^C. \quad (13)$$

The parameters of this fit are calibrated using N -body simulations and are given by: $A = 5.71$, $B = -0.084$, $C = -0.47$, and $M_{\text{pivot}} = 2 \times 10^{12} h^{-1} M_{\odot}$.

The NFW profile used in this work, in combination with the Duffy concentration relation, is illustrated in Fig. 3.

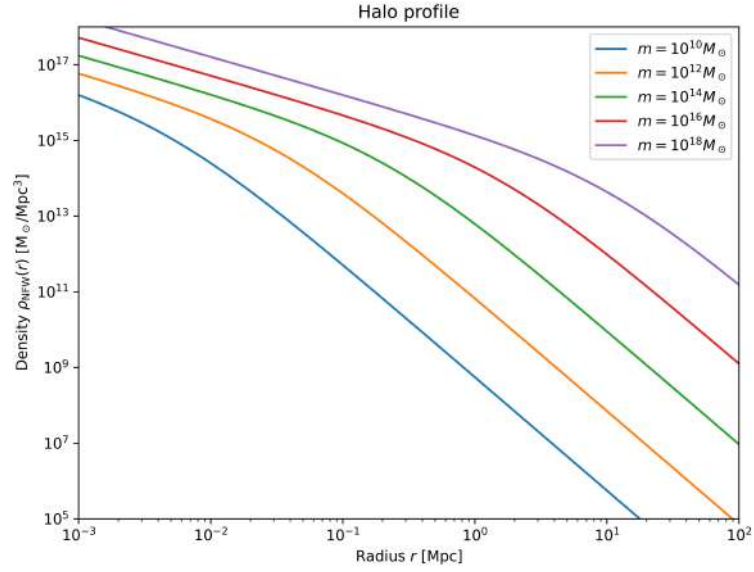


Figure 3: The NFW halo profile with the Duffy concentration relation defines the distribution of matter and galaxies within a halo of mass m at distance r from the halo center.

2.1.5 Linear Matter Power Spectrum

The linear matter power spectrum quantifies the amplitude of density fluctuations in the early Universe as a function of scale during the linear regime of structure formation. It reflects the initial conditions set by primordial fluctuations and their evolution under gravity, influenced by components such as dark matter and baryons, as well as radiation, neutrinos, and the effects of dark energy on the expansion history.

The shape and amplitude of the linear power spectrum are determined by solving the coupled Boltzmann equations that describe the evolution of perturbations in different matter and radiation components. Numerical Boltzmann codes, such as CAMB, provide highly accurate transfer functions that incorporate these physical processes (Eisenstein & Hu 1998).

2.1.6 Non-linear Power Spectrum

The non-linear power spectrum provides a statistical description of the large-scale structure of the Universe by quantifying how density fluctuations vary with spatial scale. It is defined as the Fourier transform of the two-point correlation function. For two fields $\theta_u(\mathbf{x})$ and $\theta_v(\mathbf{x}')$, this correlation function is given by

$$\langle \theta_u(\mathbf{x}) \theta_v(\mathbf{x}') \rangle = \xi_{uv}(|\mathbf{x} - \mathbf{x}'|), \quad (14)$$

where ξ_{uv} describes the spatial correlation between the fields u and v at a separation $|\mathbf{x} - \mathbf{x}'|$. In Fourier space, this relation takes the form of the power spectrum:

$$P_{uv}(\mathbf{k}) = \langle \hat{\theta}_u(\mathbf{k}) \hat{\theta}_v^*(\mathbf{k}') \rangle = \left\langle \sum_{i,j} e^{-i\mathbf{k} \cdot \mathbf{x}_i} e^{i\mathbf{k}' \cdot \mathbf{x}_j} N_i N_j \hat{W}_{u,i}(M, \mathbf{k}) \hat{W}_{v,j}(M, \mathbf{k}') \right\rangle, \quad (15)$$

where $\hat{\theta}_u(\mathbf{k})$ and $\hat{\theta}_v^*(\mathbf{k}')$ are the Fourier transforms of the respective fields, $\mathbf{x}_i$ and $\mathbf{x}_j$ are the positions of halos, N_i and N_j the number of tracers within those halos, and $\hat{W}_{u,i}(M, \mathbf{k})$ and $\hat{W}_{v,j}(M, \mathbf{k}')$ the Fourier-transformed spatial profiles (in our case NFW profiles) of the fields in halos of mass M .

2 Theoretical Background

In the context of the halo model, the power spectrum can be separated into two contributions: the one-halo term and the two-halo term. The one-halo term describes correlations of tracers within the same halo and thus dominates on small, non-linear scales. In contrast, the two-halo term accounts for correlations between tracers in different halos and dominates on larger scales. This decomposition enables a detailed and physically motivated modeling of the matter distribution across all relevant scales, which is illustrated in Fig. 4.

For two distinct fields, such as the galaxy and matter density fields, the galaxy–matter power spectrum quantifies the cross-correlation between galaxy and matter density fluctuations in Fourier space. In contrast, the matter–matter power spectrum describes the auto-correlation, i.e., the variance of matter density fluctuations across scales. Overall, the power spectrum is a fundamental tool in cosmology, as comparing its theoretical predictions with observational data allows for tighter constraints on cosmological parameters (Asgari et al. 2023).

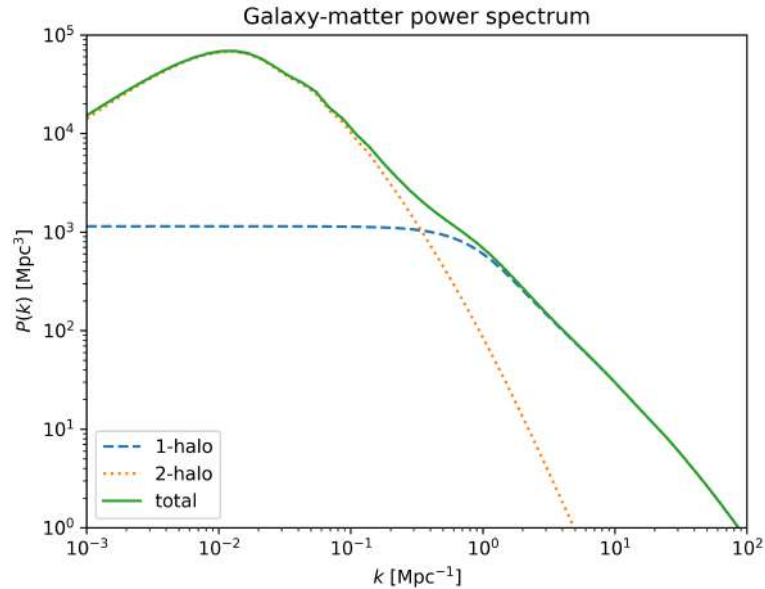


Figure 4: The non-linear galaxy-matter power spectrum describes density fluctuations in Fourier space at scales given by the wavenumber k . Additionally, the 1- and 2-halo terms composing the power spectrum are displayed.

2.1.7 Halo Occupation Distribution

The halo occupation distribution (HOD) extends the traditional halo model by specifying the expected number of galaxies residing within a dark matter halo as a function of its mass. While the halo model primarily describes the distribution of dark matter, the HOD incorporates the galaxy population, providing a more detailed and realistic connection between dark matter halos and galaxies. This extension improves upon the original halo model by enabling a more accurate representation of galaxy clustering within the cosmic large-scale structure, which is essential for studies of galaxy evolution, refined large-scale structure statistics, and more precise cosmological parameter inference (Linke et al. 2022).

The HOD model used in this work is based on the framework proposed by Zheng et al. (2007), which differentiates between central and satellite galaxies. For both galaxy types, the model specifies the expected number of galaxies per halo, leading to the following expression for the total number of galaxies in a halo of mass m :

$$\langle N \mid m \rangle = \langle N_{\text{cen}} \mid m \rangle + \langle N_{\text{sat}} \mid m \rangle \quad (16)$$

Here, $\langle N \mid m \rangle$ denotes the mean number of galaxies in a halo of mass m , composed of $\langle N_{\text{cen}} \mid m \rangle$, the mean number of central galaxies, and $\langle N_{\text{sat}} \mid m \rangle$, the mean number of satellite galaxies.

Additionally restrictions of the model state that each halo can at most host one central galaxy $N_{\text{cen}} \in [0; 1]$, located at the halo center but can host several satellite galaxies $N_{\text{sat}} \in [0; \infty)$. The mean number of central galaxies $\langle N_{\text{cen}} \mid m \rangle$ is only dependent on the halo mass m , similar to the approach of Zheng et al. (2007),

$$\langle N_{\text{cen}} \mid m \rangle = \frac{\alpha}{2} \left[1 + \text{erf} \left(\frac{\log(m) - \log(M_{\text{th}})}{\sigma} \right) \right]. \quad (17)$$

The free parameters α , M_{th} and σ can be calibrated to fit simulations and observational data. M_{th} describes the threshold mass below no galaxies are expected in

2 Theoretical Background

a halo. The parameter α gives the maximum mean number of central galaxies for massive halos ($m \gg M_{\text{th}}$), while σ describes the smoothness of the transition from $\langle N_{\text{cen}} | m \rangle = 0$ to $\langle N_{\text{cen}} | m \rangle = \alpha$ (Linke et al. 2022).

The mean number of satellite galaxies $\langle N_{\text{sat}} | m \rangle$ is described following the approach of Zehavi et al. (2005),

$$\langle N_{\text{sat}} | m \rangle = \frac{1}{2} \left[1 + \text{erf} \left(\frac{\log(m) - \log(M_{\text{th}})}{\sigma} \right) \right] \left(\frac{m}{M'} \right)^\beta. \quad (18)$$

The free parameters M' and β characterize the mass dependence of satellite galaxy occupation. The parameter M' sets the mass scale at which satellite galaxies begin to appear in significant numbers, while β controls how quickly the number of satellites increases with halo mass. Together with M_{th} and σ , which shape the central galaxy component, these parameters allow modeling the observed occurrence and distribution of satellite galaxies across different halo masses. As illustrated in Fig.5, the satellite occupation follows the central galaxy occupation at low halo masses and transitions into a power-law behavior at higher masses (Linke et al. 2022).

2 Theoretical Background

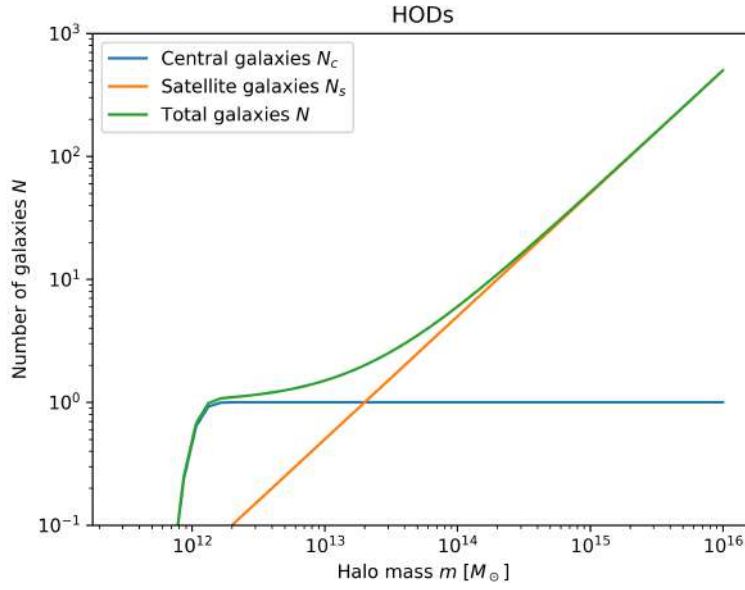


Figure 5: The HOD describes the expected number of galaxies residing within a halo of mass m . The sharp decline of galaxies for low-mass halos results from the threshold mass M_{th} , below which no galaxies are expected in a halo (Laila Linke, personal communication, 2025).

2.2 Neural Networks

In this work, computationally intensive numerical models of large-scale structure statistics will be accelerated using neural networks trained on a regression task, learning surrogate models that approximate the underlying physical processes.

Neural Networks (NNs) are a class of machine learning architectures inspired by the structure and functioning of the human brain. They are designed to model complex relationships and identify patterns within data. Due to their high efficiency in approximating any arbitrary unknown functions based on a set of input-output samples, NNs have become a state-of-the-art method in various scientific domains (Zhang et al. 2021, Schmidhuber 2015). Especially in cases where no exact theoretical model is available or where theoretical models are difficult to compute analytically, large datasets can be utilized for training these neural networks (Basheer & Hajmeer 2001).

2.2.1 Surrogate Models

Modeling real physical systems often involves a large number of input parameters and requires computationally expensive calculations. Surrogate models offer an efficient solution to this problem by learning a mapping from the input space to the output space, thereby approximating the underlying functional relationship with significantly reduced computational cost. Especially for high-dimensional input spaces, NNs act as powerful surrogate models by capturing complex, nonlinear dependencies within the data. By appropriately designing the network architecture, the surrogate model can be interpreted as recovering a low-dimensional, nonlinear manifold embedded in the high-dimensional input space. The model response is then expressed as a parameterized nonlinear function of these low-dimensional projections. This perspective not only reduces the effective dimensionality of the problem but also enhances generalization and data efficiency by leveraging the intrinsic structure of the system’s behavior (Tripathy & Bilonis 2018).

2.2.2 Architecture

The fundamental architecture of a conventional NN consists of a large number of simple, interconnected processing units, commonly referred to as neurons. Each neuron produces a sequence of real-valued activations. These neurons are organized into layers: an input layer and an output layer, which correspond to the dimensionality of the input and output data, respectively, as well as one or more hidden layers containing varying numbers of neurons. The hidden layers enable the network to learn complex nonlinear mappings between inputs and outputs. Information is propagated through the network via weighted connections between neurons, where each neuron is connected to all neurons in the preceding layer. These weights encode the relative importance of transmitted information and are adjusted during the training process (Section 2.2.4) to optimize the network's behavior toward a desired output (Schmidhuber 2015, Basheer & Hajmeer 2001).

A schematic example illustration of the NN architecture is shown in Fig. 6.

The value of a neuron in a given layer is computed as a weighted sum of the activations from the previous layer. These activations, denoted by a_n , serve as the output signals of the preceding neurons. Formally, the pre-activation input x_k to neuron k is given by

$$x_k = \sum_n w_{k,n} a_n . \quad (19)$$

Here, $w_{k,n}$ denotes the weight of the connection from neuron n in the preceding layer to neuron k in the current layer. The resulting value x_k is then passed through a non-linear activation function f , yielding the activation a_k of the neuron,

$$a_k = f(x_k) . \quad (20)$$

The activation function f introduces non-linearity into the network, which is essential for learning complex, non-linear mappings between inputs and outputs

2 Theoretical Background

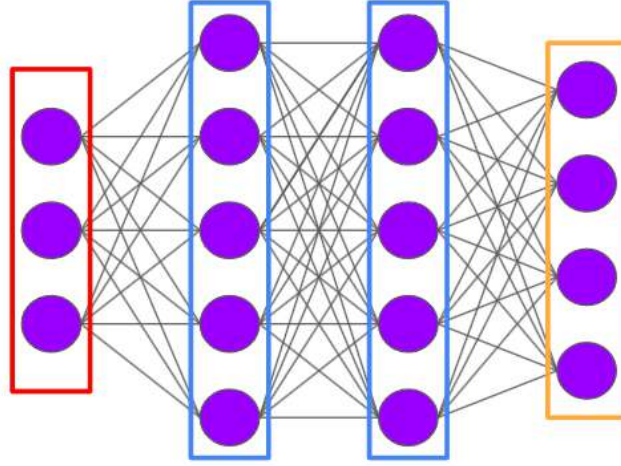


Figure 6: An example architecture of a Neural Network is shown. Purple represents the neurons of the network, colored boxes highlight different layer, where red is the input layer, blue are the hidden layers and orange is the output layer. The grey connections illustrate the weighted connections between the neurons.

(Schmidhuber 2015). Commonly used activation functions include the sigmoid and the Rectified Linear Unit (ReLU), as illustrated in Fig. 7.

This architecture forms the foundation for the computational efficiency of neural networks, as discussed in Section 2.2.6, since all computations within the network are based solely on matrix and vector multiplications (Schmidhuber 2015) of the form $f(W \cdot \mathbf{a}^{(l)}) = \mathbf{a}^{(l+1)}$,

$$f \left(\begin{bmatrix} w_{0,0} & w_{0,1} & \cdots & w_{0,n} \\ w_{1,0} & w_{1,1} & \cdots & w_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{k,0} & w_{k,1} & \cdots & w_{k,n} \end{bmatrix} \cdot \begin{bmatrix} a_0^{(0)} \\ a_1^{(0)} \\ \vdots \\ a_n^{(0)} \end{bmatrix} \right) = \begin{bmatrix} a_0^{(1)} \\ a_1^{(1)} \\ \vdots \\ a_k^{(1)} \end{bmatrix}. \quad (21)$$

In this expression, w refers to elements of the weight matrix, while $\mathbf{a}$ describes the activation vector of layer l .

2 Theoretical Background

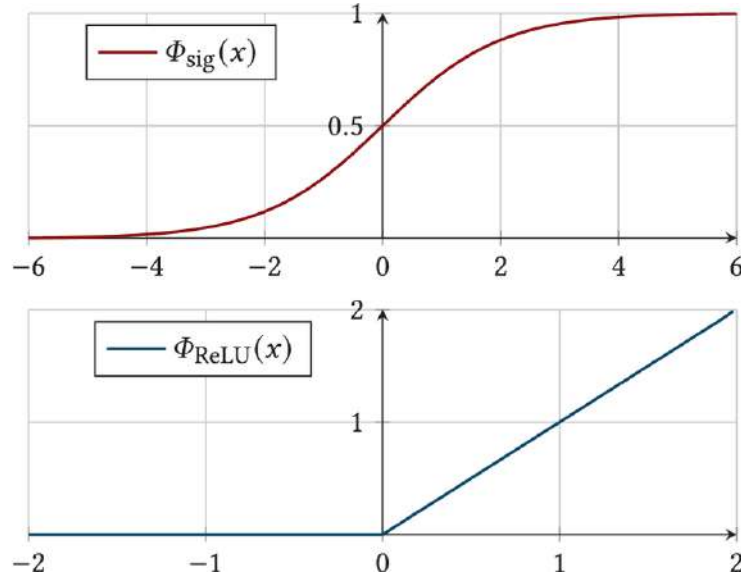


Figure 7: Two standard activation functions of neural networks are shown. On the top the Sigmoid function is displayed and on the bottom the ReLU function (Stock et al. 2022).

2.2.3 Learning

The learning process of a neural network fundamentally involves adapting the weight parameters of the network architecture to produce the most accurate predictions across all provided data. The accuracy of a prediction for a given data sample is quantified by the mean squared error (MSE),

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - t_i)^2. \quad (22)$$

Here, y_i denotes the predicted value and t_i the target (true) value for the i -th data point of a sample consisting of n points. In the context of machine learning and in the following, this quantity is referred to as the *loss*.

The loss is a function dependent on all network weights, i.e., $L(w_{0,0}, \dots, w_{k,n})$, and forms a complex, high-dimensional error landscape. To find the optimal weight configuration for the given problem, the loss is summed over all training samples, and the goal is to find the global minimum of this aggregated loss function. This mini-

2 Theoretical Background

mization is performed iteratively using a modified gradient descent algorithm, which is determined by the chosen optimizer—a key hyperparameter of neural networks (Basheer & Hajmeer 2001).

Gradient Descent

Gradient descent is an optimization algorithm for functions in high-dimensional spaces that iteratively moves closer to a local minimum. At each iteration, the gradient of the function at the current point is computed; this gradient indicates the direction of steepest ascent. The algorithm then updates the point by moving in the opposite direction of the gradient, thereby approaching a local minimum (Goodfellow et al. 2016).

In the context of neural networks, a point in the loss landscape corresponds to a specific set of weights and the associated loss value. The weight update at iteration j is given by

$$w_{j+1} := w_j - \frac{\partial \text{MSE}}{\partial w_j}. \quad (23)$$

Here, w_j denotes the weight at the current iteration, w_{j+1} the updated weight for the next iteration, $\partial \text{MSE} / \partial w_j$ the partial derivative (gradient) of the loss with respect to w_j (Basheer & Hajmeer 2001).

Over the past decades, the basic gradient descent algorithm has been enhanced with various modifications to improve efficiency in neural network optimization. These enhancements include techniques such as *momentum*, which accelerates convergence, and adaptive learning rates to avoid getting stuck in local minima or overshooting minima. Today, many such optimizers exist, each tailored to different types of problems. One widely used example is the Adam optimizer (Kingma & Ba 2017).

2.2.4 Training

Training a neural network requires a sufficiently large dataset consisting of input–output (feature–target) pairs. The network learns from this dataset over many iterations, referred to as epochs. During training, the dataset is divided into smaller subsets called batches. The training process begins with randomly initialized network weights. For each batch, the network performs a forward pass, generating predictions based on the current weights. The quality of the predictions is then calculated, and the resulting loss is used to compute gradients via a backward pass (backpropagation). The gradients are then applied to update the weights through an optimization algorithm.

To reduce the risk of overfitting, regularization techniques such as dropout are often employed during training. Dropout randomly deactivates a fraction of neurons in the network on each forward pass, forcing the model to learn more robust and generalizable features by preventing co-adaptation of neurons. This stochastic neuron removal effectively trains an ensemble of subnetworks, improving the model’s ability to generalize to unseen data.

After completing one pass through all batches (i.e., one epoch), the network performs a forward pass on a separate validation dataset, which was not used during training. Monitoring the validation loss provides insight into the model’s generalization performance and helps detect overfitting, while also assessing convergence behavior. This process is repeated over many epochs, ideally until the validation loss stabilizes and convergence is achieved.

2 Theoretical Background

Backpropagation

In order to optimize the weights of a neural network using gradient descent, we must compute the gradient of the loss function L with respect to each weight in the network. This is done via the *backward pass*, also known as backpropagation.

The backward pass begins with the output layer. Assuming a differentiable activation function f , the output of a neuron k in the output layer is given by $y_k = f(x_k)$, where x_k is the total input to the neuron.

Applying the chain rule, we compute the derivative of the loss with respect to the neuron's input,

$$\frac{\partial L}{\partial x_k} = \frac{\partial L}{\partial y_k} \frac{\partial y_k}{\partial x_k} = (y_k - t_k) f'(x_k). \quad (24)$$

The total input x_k is a linear combination of the activations a_n from the previous layer n , as defined in Eq. 19. Using this, the gradient with respect to the weight $w_{k,n}$, connecting neuron n to neuron k , becomes

$$\frac{\partial L}{\partial w_{nk}} = \frac{\partial L}{\partial x_k} \frac{\partial x_k}{\partial w_{nk}} = (y_k - t_k) f'(x_k) a_n. \quad (25)$$

This completes the gradient computation for the output layer and forms the basis for recursively propagating the error backward through the hidden layers. The error at a hidden unit n is determined by the weighted sum of the backpropagated errors from units in the next layer k ,

$$\frac{\partial L}{\partial a_n} = \sum_k \left(\frac{\partial L}{\partial x_k} w_{nk} \right). \quad (26)$$

This process is applied recursively from the output layer back to the earliest hidden layer, enabling the computation of all necessary gradients for updating the network weights (Rumelhart et al. 1986).

2.2.5 Hyperparameter Optimization

In addition to the inner optimization problem inherent to neural networks, which aims to find the optimal weight configuration, there exists an outer optimization problem. The architecture of the neural network, as described in Section 2.2.2, has a significant impact on the model’s performance. It determines the total number of weights through the number of layers and neurons per layer, thereby directly influencing the inner optimization problem by shaping the loss function landscape. Moreover, it affects the training process by defining the choice of optimizer algorithm, activation function, learning rate, batch size, and the number of training epochs. These factors influence both the fundamental learning dynamics and the convergence behavior during training, ultimately determining which minima of the loss function are reached. Consequently, the objective of the outer optimization problem is to identify the architecture that yields the best weight configuration, producing the most accurate predictions.

Since the outer optimization problem inherently involves the inner optimization, it cannot be solved independently. Therefore, neural networks must be trained under various hyperparameter configurations, and their performance evaluated accordingly. While this might suggest that a grid search over the hyperparameter space would suffice, Bergstra & Bengio (2012) demonstrated that random search outperforms grid search in the majority of hyperparameter optimization tasks. This result is counterintuitive from a human perspective but highlights the inefficiency of grid search, especially given the substantial computational cost associated with training numerous neural network models.

Bayesian Optimization

A new and more efficient method for hyperparameter optimization, suggested by Snoek et al. (2012), is Bayesian optimization. This approach places a Gaussian process (GP) prior over the unknown objective function in the hyperparameter space, thereby modeling our belief about its behavior before any observations are made. Each new observation, i.e., training a model with a specific set of hyperparameters and evaluating its performance, updates the posterior distribution of the GP, refining our estimate of the function and quantifying uncertainty in unexplored regions. The posterior thus captures both the most likely function values and the confidence in those estimates. Based on this updated model, the algorithm selects the next configuration to evaluate by optimizing an acquisition function, such as the Expected Improvement, Probability of Improvement, or Upper Confidence Bound, which balances exploration of uncertain regions and exploitation of promising areas. Importantly, Bayesian optimization is not limited to refining known minima, as it is explicitly designed to explore the broader hyperparameter landscape in order to discover previously unknown optima that may yield superior performance. As a result, each training run contributes new, targeted insights that help guide the search more effectively, enabling continuous and computational-efficient improvement of the model.

2.2.6 Predictive Modeling for Computational Efficiency

As discussed in Section 2.2.1, neural networks offer an efficient approach to approximate results that would otherwise require computationally expensive model evaluations. By learning to map high-dimensional input spaces to corresponding output spaces through surrogate modeling, neural networks can generate predictions several orders of magnitude faster than traditional numerical methods.

The foundation of this efficiency lies in the architecture of neural networks, as described in Section 2.2.2. The prediction process in a trained neural network is

2 Theoretical Background

based on matrix operations, which are highly optimized for parallel computation on modern hardware architectures such as GPUs or TPUs. In contrast, the evaluation of physical models or simulations typically involves computationally intensive steps such as solving systems of high-dimensional differential equations, performing integrations, or executing iterative solvers.

Once trained, a neural network can approximate outputs using only matrix multiplications and activation functions, making the prediction process highly efficient. Furthermore, neural networks support parallel batch evaluation, allowing simultaneous predictions for a large number of input samples, which further enhances computational throughput (Goodfellow et al. 2016). Traditional calculations of physical models, by contrast, require the full computational cost to be incurred separately for each individual evaluation.

A prominent example closely related to the problem addressed in this work is the work by Spurio Mancini et al. (2022), in which neural networks are used to accelerate cosmological matter power spectrum predictions. In addition, similar approaches have been successfully applied in various scientific and engineering contexts, such as surrogate modeling of gravitational waveforms (Field et al. 2014), accelerating physics-based simulations in oil reservoir modeling (Navrátil et al. 2019), solving diffusion equations (Toledo-Marín et al. 2021), filling gaps in satellite ocean color data (Krasnopolsky et al. 2015), and building versatile high-accuracy emulators across diverse fields using neural architecture search (Kasim et al. 2021).

3 Realization and Implementation

The following chapter describes the process of training various neural networks in order to find the best architecture for predicting power spectra implemented in Python. In the following, this process is illustrated using the galaxy-matter power spectrum as an example; however, the same procedure was also carried out for the galaxy-galaxy and matter-matter power spectra.

3.1 Data Generation

As mentioned in Section 2.2.4, training a neural network requires a sufficiently large dataset of feature–target samples on which the model can learn. Therefore, the first step was to generate a dataset of appropriate size by computing the power spectra corresponding to different cosmological configurations using the HOD-Halo model implementation by Linke et al. (2022). For a given cosmological configuration, defined by 10 parameters (5 cosmological parameters and 5 HOD parameters), this model computes the 1-halo, 2-halo, and total term of the power spectrum, represented by 50 function values over different scales of the wavenumber $k \in [10^{-2}, 10^2]$.

The cosmological and HOD parameter priors used for sampling are summarized in Table 3.1. These priors are chosen such that the true parameter values are expected to lie within them with high confidence, based on the measurements of Linke et al. (2022).

A dataset comprising 1,000,000 samples was generated for the purpose of training,

3 Realization and Implementation

validating, and testing the models. While the generation of such a large dataset is computationally expensive, it is a crucial prerequisite for enabling effective model training and for conducting a reliable performance evaluation.

Another important aspect of the dataset is the homogeneous distribution of data samples across the feature parameter space. This ensures that the neural network can learn the complex relationships between features and targets uniformly throughout the entire parameter space, avoiding regions that are over- or underrepresented during training. To achieve this, the dataset was generated using Latin Hypercube Sampling (LHS), which guarantees approximately uniform sampling density within each subvolume of the parameter space. The distribution of the training samples in the parameter space is illustrated in Fig. 8.

Table 3.1: Intervals of the cosmological and HOD parameters used for data generation.

Cosmological Parameters		HOD Parameters	
Parameter	Interval	Parameter	Interval
Ω_c	[0.1, 0.6]	α	[0, 1]
Ω_b	[0.04, 0.06]	σ	[0.01, 1]
h	[0.64, 0.82]	M_{th}	$[10^{10} M_\odot, 10^{15} M_\odot]$
σ_8	[0.8, 1.0]	M'	$[10^{10} M_\odot, 10^{17} M_\odot]$
n_s	[0.84, 1.1]	β	[0.1, 2]

3.2 Data Preprocessing

Several preprocessing steps are necessary before the dataset can be used to train a neural network. First, the data must be split into training, validation, and test subsets. The training set is used to optimize the model weights, while the validation set serves to monitor performance on unseen data during training, as detailed in Section 2.2.4. This enables hyperparameter tuning and helps prevent overfitting.

3 Realization and Implementation

The test set provides an unbiased estimate of the final model performance after training is complete.

In our case, the dataset was split into 50,000 training samples, 10,000 validation samples, and 940,000 test samples. The size of the training set was chosen as a compromise between computational efficiency and the need for a sufficiently large and representative training set. While increasing the training set size generally improves model performance, it also leads to longer training times per epoch due to the increased volume of data. It is therefore important to balance these factors to ensure effective and efficient training. The large test set was chosen to enable a robust and statistically reliable evaluation of the model’s predictive performance across a universal range of data samples.

Overall, the adopted strategy was to first identify an optimal neural network architecture for predicting power spectra and then train this model on an expanded dataset to maximize predictive accuracy. Further details on this procedure are provided in Section 3.4.

Subsequently, a cross-check between these subsets is performed to ensure consistent distributions of input features and target values (power spectra) across all splits, which is essential for reliable performance evaluation. This verification is conducted by comparing the means and standard deviations of each subset, as shown in Figs. 9a and 9b. The plots confirm matching means and standard deviations across all subsets. Finally, the data is rescaled to mitigate biases resulting from features with varying orders of magnitude. The rescaling procedure normalizes all input features and target variables to have zero mean and unit standard deviation, by applying

$$x_{\text{rescaled}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (27)$$

where x denotes the original (unscaled) value, μ_{train} is the mean of the corresponding feature or target variable computed over the training subset, and σ_{train} is the respective standard deviation, also calculated from the training subset.

3.3 Neural Network Training

The trained setup consists of a single neural network that receives ten rescaled cosmological input parameters and predicts 150 function values of the power spectrum. These predicted values are subsequently divided into 50 values each for the 1-halo, 2-halo, and total terms. In order to identify an optimal model architecture for this setup, a large number of neural networks were trained using hyperparameter optimization.

As detailed in Section 2.2.5, this optimization was performed efficiently using Bayesian optimization. A Bayesian optimization algorithm was implemented with the **Optuna** library (Akiba et al. 2019) to automatically train successive models, continuously improving them by monitoring the test loss. Neural network training was conducted using the **TensorFlow** framework (Abadi et al. 2015). All trained models, along with their corresponding hyperparameters, were stored to enable subsequent evaluation and ensure reproducibility.

To constrain the hyperparameter space, each hyperparameter was assigned a fixed interval within which the search was conducted. These ranges are summarized in Table 3.2. The choice of optimizers was restricted to **Adam** and **RAdam**, as these optimizers exhibited the best performance in a preliminary Bayesian optimization study, as illustrated in Fig. 11.

Each network was trained in three fixed learning steps, a common practice in machine learning that ensures comparability between models. Specifically, the model was trained for 1000 epochs, then another 1000 epochs, and finally 10,000 epochs, with the learning rate reduced by a factor of 10 after each step. Early stopping was applied during training to prevent overfitting by monitoring the validation loss and stop training once convergence was achieved. After training, the model’s weights were restored to those from the epoch with the best validation performance. This approach not only reduces the overall training time and hyperparameter complexity but also allows the model to first learn coarse structures in the dataset, which are

3 Realization and Implementation

Table 3.2: Hyperparameter search intervals used for Bayesian optimization.

Hyperparameter	Range / Options
Number of Layers	[1, 20]
Neurons per Layer	[16, 2048]
Learning Rate	$[10^{-6}, 10^{-2}]$
Activation Functions	relu, selu, leaky_relu, elu, gelu, swish, sigmoid, tanh
Optimizers	adam, RAdam
Batch Size	[16, 2048]
Dropout	[0, 0.5]

subsequently fine-tuned in later stages using the reduced learning rate. This staged training procedure is illustrated in Figure 10.

After training, the models were evaluated based on their final test loss. The best-performing models were subsequently subjected to a more detailed analysis using the test dataset in order to assess their predictive performance across the entire range of the power spectrum (see Section 4.1).

To explore an alternative neural network architecture for predicting power spectra, a second setup was also trained using Bayesian optimization. In this configuration, two separate models were trained, each responsible for predicting 50 power spectrum function values corresponding to the 1-halo and 2-halo terms, respectively. The total term was then computed by summing the predictions of both models. This design allows each network to specialize in learning one specific term, potentially improving overall prediction accuracy. A comparison between the two setups is presented in Section 4.1 and 4.2.

3.4 Neural Network Finetuning

Finetuning a neural network refers to the process of further improving a model’s predictive accuracy beyond the best-performing configurations identified during the initial training phase. This is typically achieved by continuing the training process with additional optimization steps. However, as shown in Fig. 10, the validation loss has already reached convergence, necessitating a change in strategy.

A first step toward improving the model involves gaining insight into its performance across the parameter space. This is done by visualizing the distribution of prediction errors with respect to the input features. Such a plot allows one to identify regions in the parameter space where the model performs particularly poorly. An example of this kind of visualization is shown in Fig. 12. From this figure, it is evident that the model exhibits systematically higher errors as $\alpha \rightarrow 0$.

To mitigate this issue, a smaller targeted dataset of 10,000 samples was generated, restricted to the subspace $\alpha \in [0, 0.1]$. The model was then further trained using this dataset, with the goal of improving its predictive performance in the critical region.

This additional training step was combined with a Bayesian optimization procedure that varied hyperparameters such as learning rate, batch size, and number of epochs in order to identify the optimal configuration. However, none of the resulting models showed an improvement; in fact, the overall predictive performance across the entire parameter space deteriorated.

3 Realization and Implementation

Subsequent attempts to merge the original training dataset with the newly generated one and retrain the model also failed to yield better results. Even training a new model from scratch on the combined training dataset did not improve overall accuracy. This decline in performance is most likely attributable to the inhomogeneous sampling of the additional and combined dataset, as discussed in Section 3.1. The disproportionate representation of samples in regions with initially poor performance effectively biases the training process, resulting in reduced accuracy across the full parameter space.

A method that proved effective involved performing a forward pass over the training data before each epoch to identify the samples with the highest prediction errors for the current model. Training was then conducted exclusively on the n_{bad} worst-performing samples in each epoch. In addition to the previously optimized hyperparameters, such as learning rate, batch size, and number of epochs, n_{bad} was introduced as an additional hyperparameter in the Bayesian optimization procedure to determine the optimal number of difficult samples to focus on during finetuning. This targeted training approach led to a significant reduction in test loss.

3 Realization and Implementation

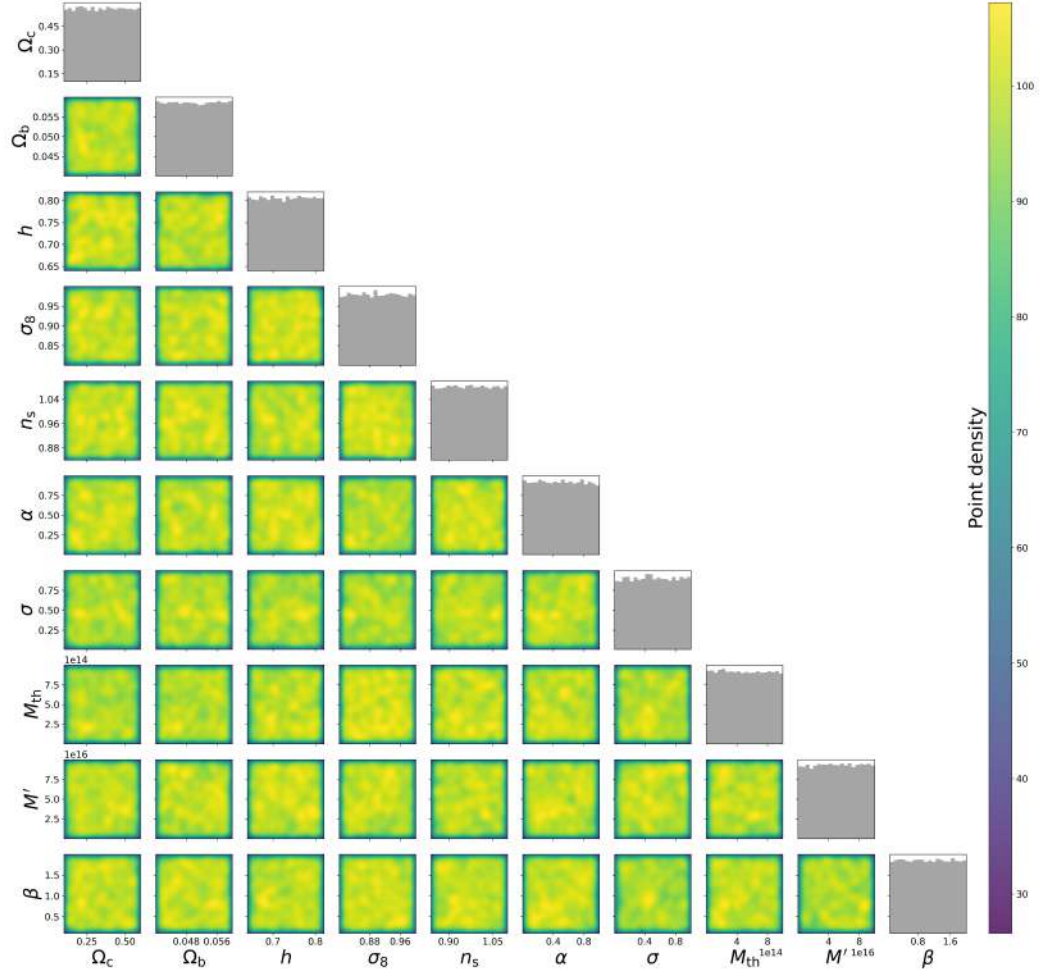
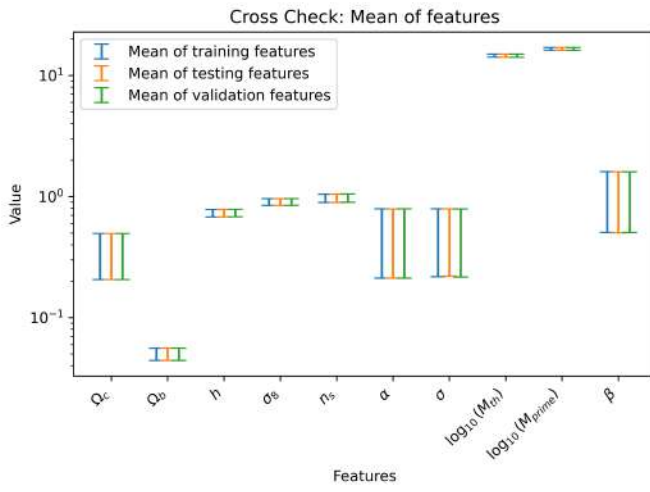
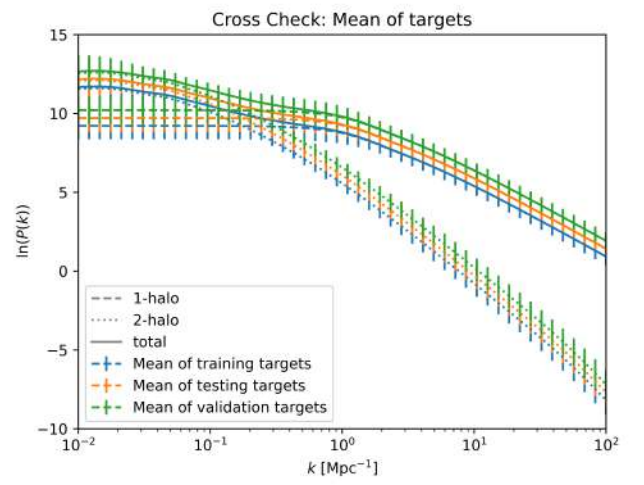


Figure 8: Data distribution of the training dataset in the parameter space achieved with LHS. The color scale expresses the point density of data samples.

3 Realization and Implementation



(a)



(b)

Figure 9: Comparison of (a) feature values and (b) target values across test, training, and validation subsets. Target values are slightly shifted for visibility, with 0.5 added to the testing targets and 1 added to the validation targets. The plots show consistent distributions of input features and target values across all splits including errors.

3 Realization and Implementation

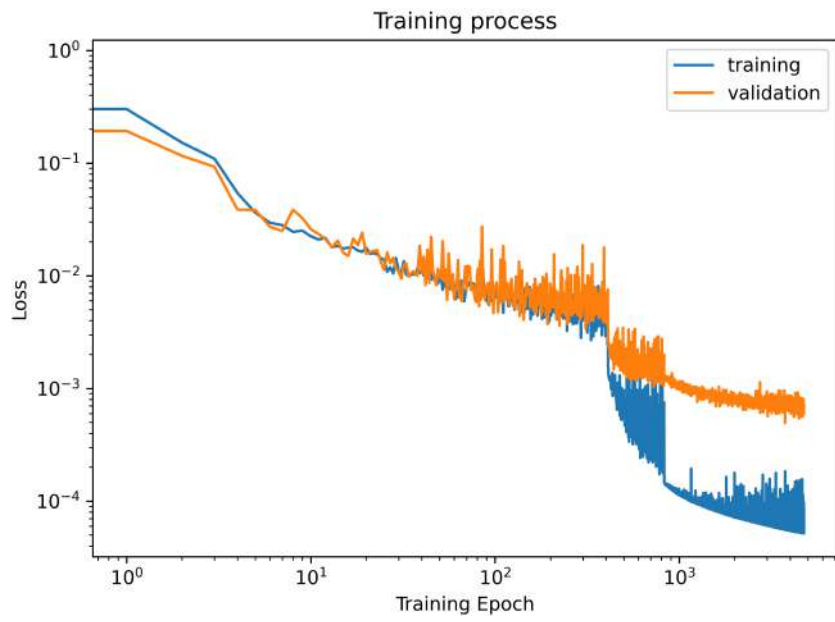


Figure 10: The evaluation of the training and validation loss during a typical training process. The training process is monitored by development of its loss and validation loss over many epochs. The transitions between the training steps are visible via the change in the convergence behavior.

3 Realization and Implementation

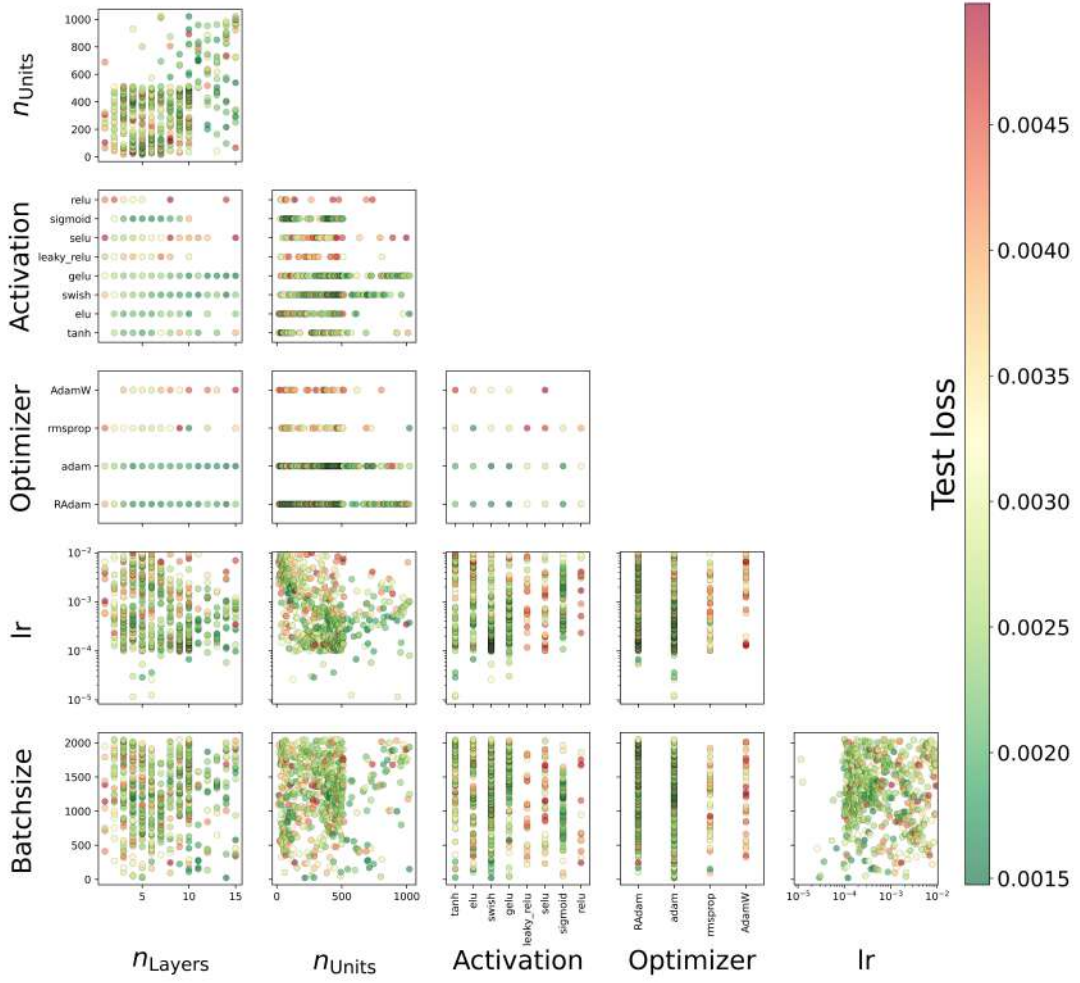


Figure 11: Shown is the hyperparameter space of a preliminary Bayesian optimization study filtered by models with a test loss lower than 0.005. After the filter there are 662 models remaining for this analysis. Each point describes the hyperparameter combination of a trained NN, with the point color describing the prediction performance over the test subset.

3 Realization and Implementation

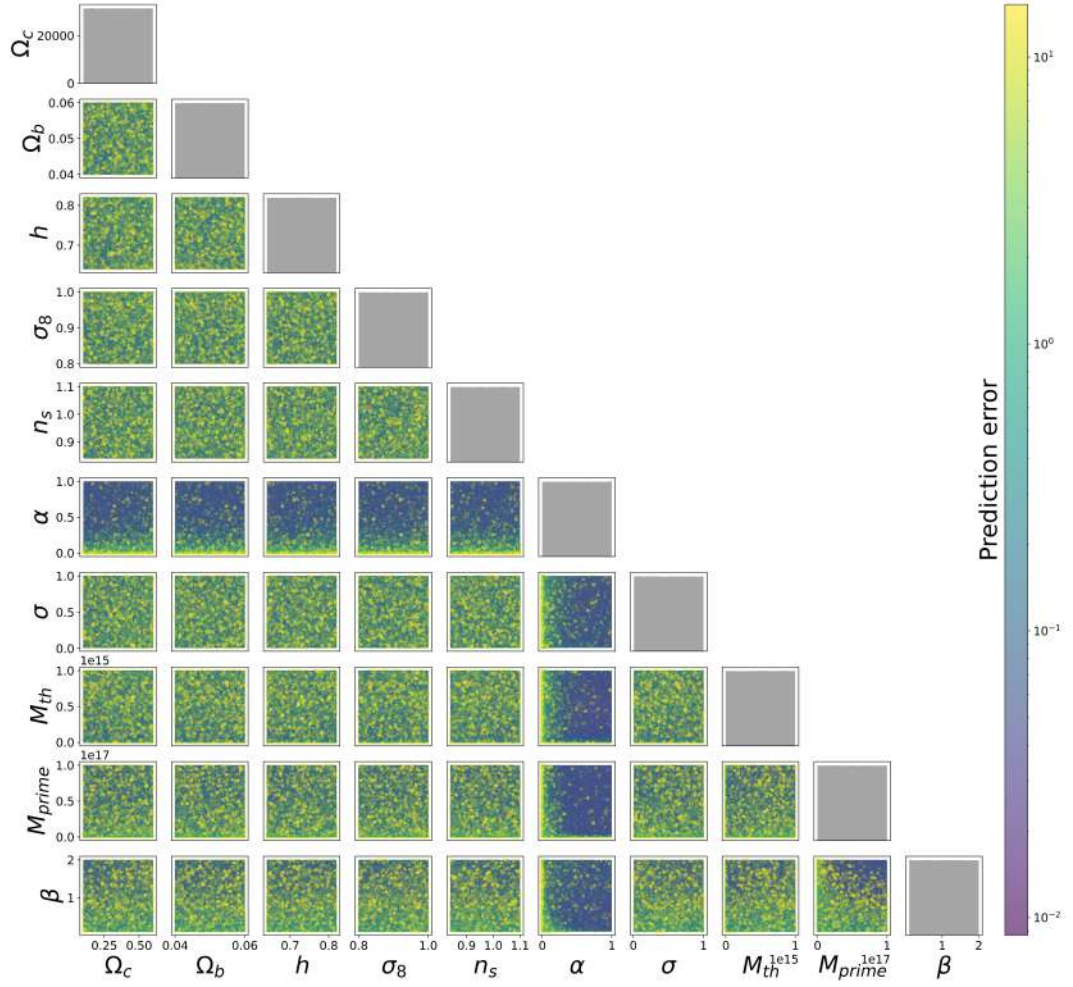


Figure 12: The parameter space with the absolute prediction error for 99.9% of samples is illustrated, showing the model performance across different regions of the parameter space.

4 Results

The following chapter presents the results of this work by comparing the predictive accuracy and computational efficiency of the trained neural network models against the exact calculations provided by the HOD-Halo model. In addition, it includes a comparison of several top-performing neural network architectures. To support transparency and reproducibility, relevant code snippets are provided in a dedicated GitHub repository, enabling others to replicate the results and apply the models to similar problems. Furthermore, an emulator is made available that allows fast and accurate predictions of power spectra based on cosmological input parameters.

4.1 Accuracy of Predictions

The accuracy of our model predictions is evaluated by comparing them to the test dataset. Figures 13a and 13b show the predicted power spectra alongside the HOD-Halo model calculations for a randomly selected sample and for the worst-predicted sample, determined by the sum of absolute errors across all data points. As illustrated, the predicted power spectra exhibit excellent agreement with the theoretical calculations.

To assess the accuracy more quantitatively, we consider the relative error between the predictions and the corresponding model values, as shown in Fig. 14a. For further insight, we analyze the relative errors separately for the 1-halo, 2-halo, and total term of the power spectrum, also depicted in Fig. 14b. Across the wavenumber

4 Results

range $k = [10^{-2}, 1] \text{ Mpc}^{-1}$, the relative error of the best model remains below 0.3% for each term in 99% of the spectra, indicating high predictive accuracy.

At $k = 10 \text{ Mpc}^{-1}$, a peak in the relative error of the 2-halo term is observed. This behavior arises because the 2-halo contribution approaches zero at these scales, causing the relative error to approach infinity. However, this is of little practical relevance, as the total signal is dominated by the 1-halo term in this regime. Of greater interest is the systematic increase in the relative error of the 1-halo term at smaller scales $k \geq 1 \text{ Mpc}^{-1}$. This effect is well-known and can be explained by the inherent numerical uncertainties of Boltzmann codes at these smaller scales (Spurio Mancini et al. 2022).

To further investigate the predictive accuracy, we compare four different model configurations and architectures. Specifically, we evaluate the relative error of each model for the 1-halo, 2-halo, and total power spectrum terms, as shown in Figures 15a, 15b, and 16. The models under comparison include a shallow (Layer ≤ 3) and a deep (Layer ≥ 10) neural network trained using the separate setup, as described in Section 3.3, as well as a shallow and a deep model trained using the combined setup, all of which achieved relatively low test losses and are therefore of potential interest.

4 Results

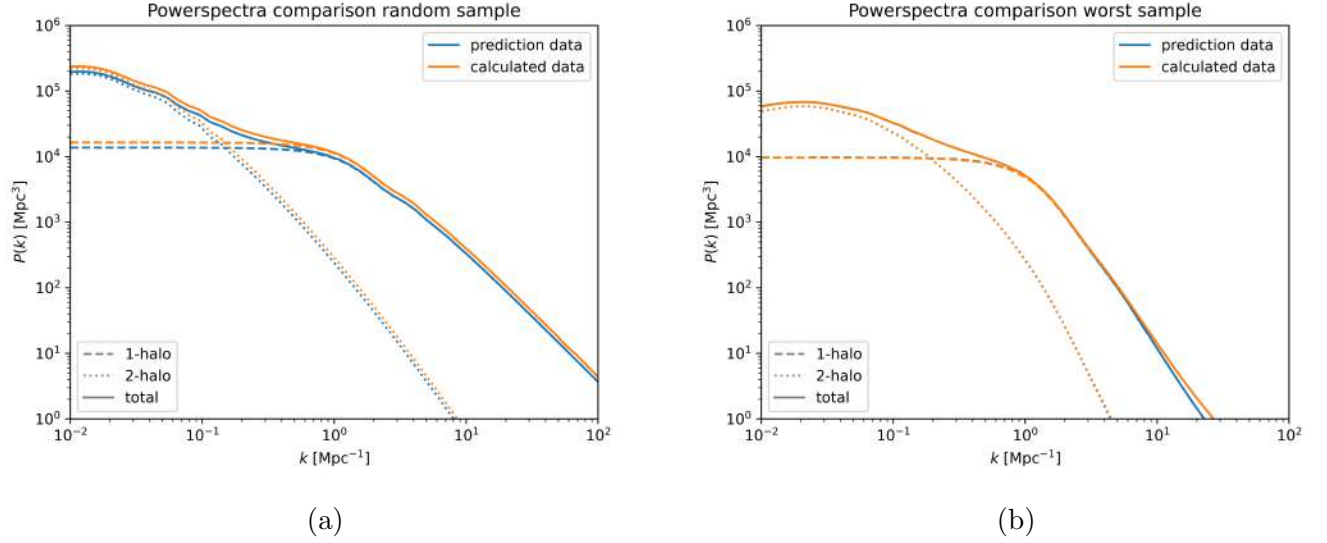


Figure 13: Prediction accuracy for a random sample (a) and the worst sample (b).

The blue line shows the predicted power spectrum and the orange line the calculated power spectrum. For the random sample, the lines are shifted by 20% for visibility.

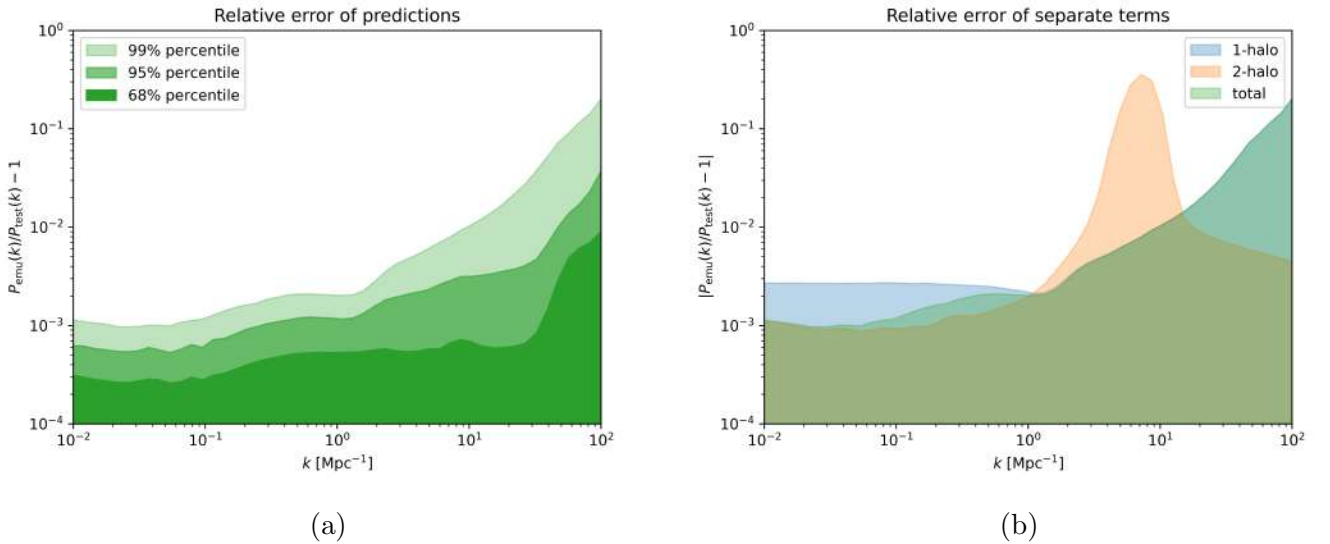


Figure 14: Relative error of predictions (a) for different percentiles of spectra, and (b) with separated prediction terms for 99% percentile.

4 Results

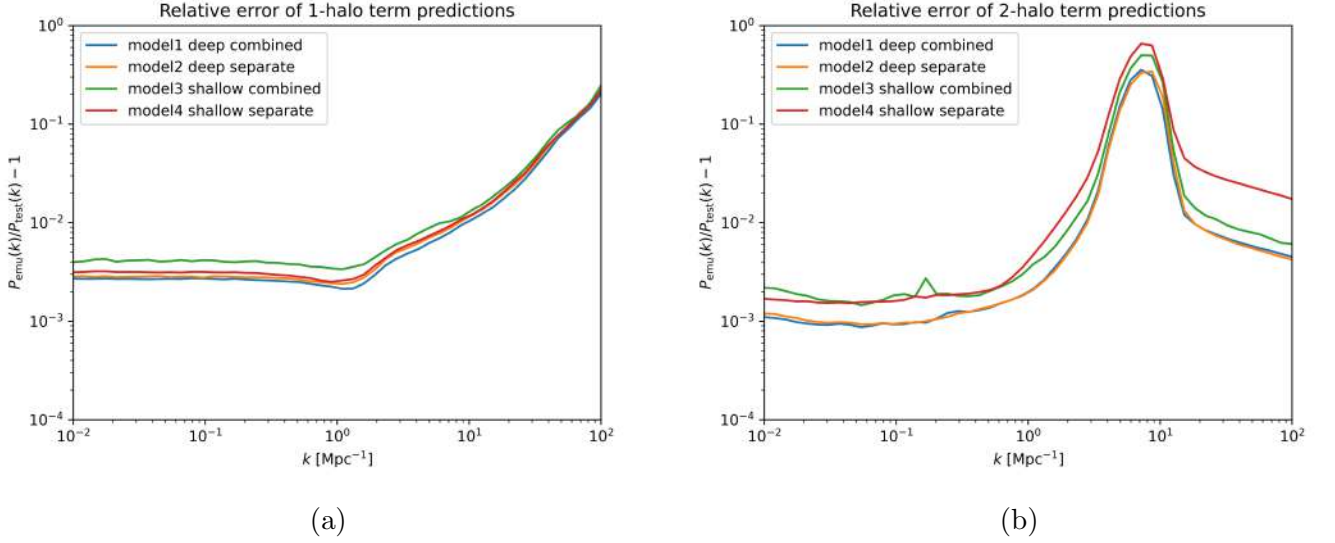


Figure 15: Comparison of the relative error for (a) the 1-halo term and (b) the 2-halo term, both showing 99% percentile of spectra for models with different neural network architectures.

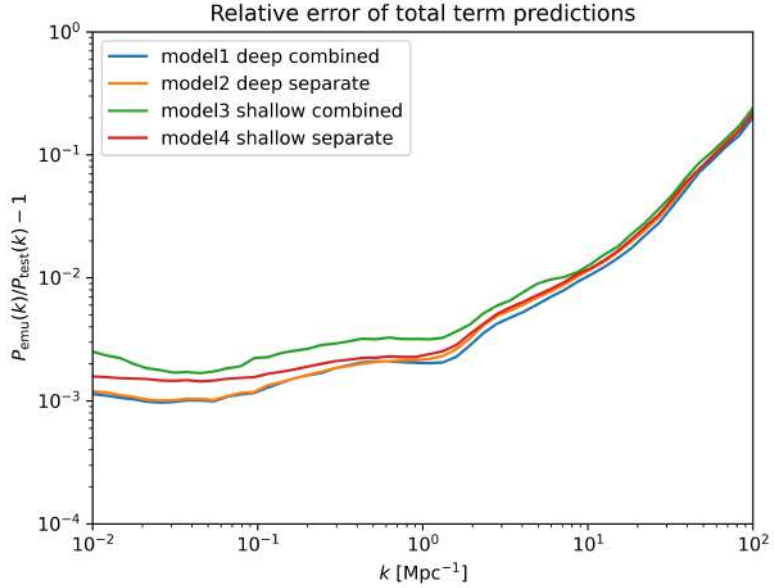


Figure 16: Comparison of the relative error for the total term including 99% percentile of spectra for models with different neural network architectures.

4.2 Performance of Predictions

To assess the computational efficiency of the best-performing model, we compare its prediction time (averaged over 10 runs) for datasets of varying size to the time required to compute the corresponding power spectra using the HOD-Halo model. This comparison is illustrated in Figure 17. For larger dataset sizes, a linear function is fitted to the halo model computation time, as the scaling behavior becomes approximately linear in this regime and the computation becomes increasingly time-consuming. The neural network predictions achieve a computational speed-up of approximately $O(10^4)$ compared to computed power spectra.

We also compare the four different model configurations and architectures in terms of prediction speed to investigate potential differences in computational performance. This comparison is illustrated in Figure 18. Additionally, we show the prediction speed against the number of weights for each model, based on the time required to generate 10 million predictions. This relation between model complexity and runtime efficiency is shown in Table 4.1. For models with separate 1-halo and 2-halo networks, both the total number of weights and the CPU time were summed to ensure a consistent comparison.

Table 4.1: Comparison neural network evaluation CPU time of 10 million samples dependent on number of weights.

Model	Architecture	Weights	CPU-Time [s]
1	deep combined	10,405,734	5620 $\pm$ 418
2	deep separate	1,925,469	1355 $\pm$ 13
3	shallow combined	535,032	210 $\pm$ 2
4	shallow separate	848,503	440 $\pm$ 5

4 Results

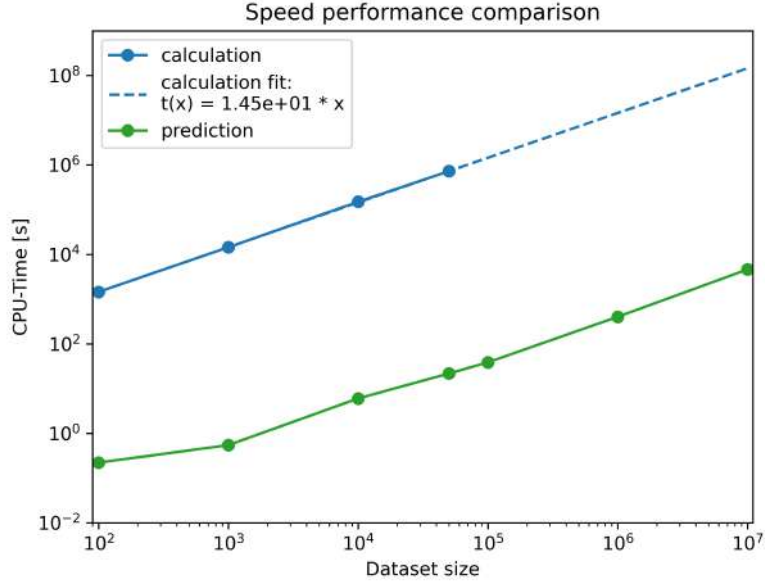


Figure 17: Comparison of the CPU-Time of calculations with the HOD-Halo model against the predictions with neural networks for different amounts of data samples.

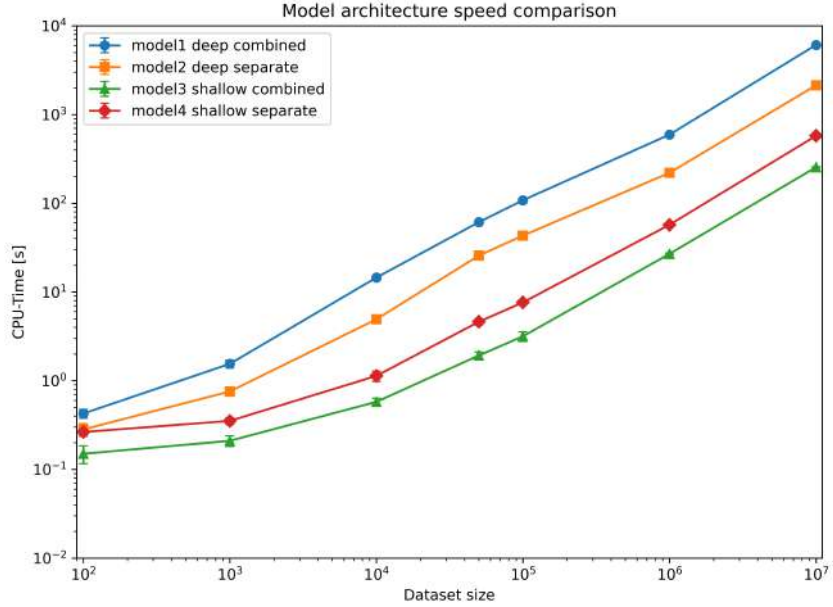


Figure 18: Comparison of the CPU-Time of models with different architectures for different amounts of data samples.

4.3 Code

All code necessary to ensure full reproducibility of the results presented in this work is provided in a dedicated GitHub repository¹. This includes comprehensive documentation and all essential components required for generating training data, preprocessing, performing Bayesian optimization (for initial training and fine-tuning of models), analyzing model quality and performance, comparing different neural network architectures, and conducting detailed dataset analyses. In addition, the repository contains the scripts for power spectrum emulation, all trained models used in this study, and the complete set of generated datasets and parameter samples. The provided framework enables users to train their own neural network models for similar use cases.

¹<https://github.com/r0b42o/NNPowerspectra>

5 Discussion

This thesis presents neural network models for predicting two-point statistics of large-scale structure, including matter–matter, galaxy–matter, and galaxy–galaxy power spectra, across a range of cosmological configurations within the HOD-halo model framework. The aim is to deliver accurate and computationally efficient predictions to accelerate the modeling process. Bayesian optimization is employed to tune the networks for this specific task, enabling effective exploration of the high-dimensional parameter space and providing fast surrogate models that can replace more expensive traditional calculations in cosmological analyses.

The results demonstrate that neural networks can predict power spectra with high accuracy for the HOD-halo model while offering orders-of-magnitude reductions in computational time. Across the wavenumber range $k = [10^{-2}, 1] \text{ Mpc}^{-1}$, the relative error of the best model remains below 0.3% for each term in 99% of the spectra.

It is important to note that the reduced accuracy observed on small scales (i.e., for $k \geq 1 \text{ Mpc}^{-1}$) likely does not stem from limitations of the neural networks themselves, but rather from the Boltzmann codes used to generate the training data (Spurio Mancini et al. 2022). Consequently, including predictions up to $k = 100$ may not be meaningful given the inherent model uncertainties. For further analysis, it is therefore advisable to restrict predictions to scales where the underlying physics is reliably modeled.

Larger training datasets improve predictive precision, although this comes at the cost of increased training time. Similarly, deeper network architectures tend to

5 Discussion

yield more accurate predictions but also require longer evaluation and training times. However, this increase remains marginal compared to the computational cost of the exact calculation of power spectra. Further setup optimization involves separating the prediction of the 1-halo and 2-halo terms. Training deep, separate networks for these components results in faster predictions while maintaining a comparable level of accuracy relative to a single, deep combined network. In practice, a balance must be struck between accuracy and efficiency: the prediction error should be significantly smaller than the statistical uncertainty introduced by observational data, beyond which faster evaluation times become preferable.

A more detailed analysis of the prediction errors across the parameter space reveals systematic inaccuracies in regions where the parameter α , which controls the mean number of central galaxies in the HOD model, approaches zero. All trained networks exhibit difficulties in these regimes, indicating that this behavior is not a limitation of the neural network approach itself, but rather a fundamental issue of the HOD-halo model. To address this, the parameter space should be refined, and the networks retrained accordingly. Such refinement may also reveal additional problematic regions where the physical model becomes unreliable, though current inaccuracies are dominated by the behavior at low α .

All predictions presented here are based on data generated over a deliberately chosen parameter space designed to span plausible ranges for all cosmological parameters in the Λ CDM and HOD-halo models. The primary motivation of this work is to enable efficient Bayesian posterior sampling using neural-network-based predictions of power spectra. In this inference framework, model predictions are used to compute the likelihood of various parameter combinations given observational data, allowing the identification of the most probable model parameters and their uncertainties. Since this approach requires generating a very large number of power spectra predictions to thoroughly explore the parameter space, the acceleration achieved through neural networks is particularly valuable (Lewis & Bridle 2002).

5 Discussion

As a first application, the parameters of the HOD-halo model will be constrained using Bayesian posterior sampling informed by data from the *Euclid* Survey — a mission designed to investigate the geometry and expansion history of the dark Universe (Euclid Collaboration et al. 2025) — which will provide a very large galaxy sample that can be employed with this work to gain deeper insight into the galaxy–matter connection. To support this and future analyses, the results of this work are made available as a user-friendly pipeline that automatically selects and applies the current best-performing neural network for power spectrum prediction. By enabling fast and accurate predictions across a wide range of HOD configurations, this approach facilitates efficient Bayesian posterior sampling of halo occupation distribution parameters, thereby improving constraints on the connection between galaxy distributions and the underlying dark matter field. This is particularly relevant in the context of upcoming large-scale surveys like *Euclid*, where precision modeling of galaxy clustering and lensing signals will be essential. The emulator developed here can directly support joint analyses of clustering, weak lensing, and galaxy abundances — similar to those performed in recent studies (Zacharegkas et al. 2025, Coupon et al. 2015, Chaurasiya et al. 2023) — by significantly accelerating the prediction of theoretical power spectra within the HOD-halo model framework. In doing so, it contributes to more detailed and statistically robust modeling of the galaxy distribution in connection with the dark matter field, thereby enabling more accurate modeling of the large-scale structure of the Universe and ultimately contributing to a deeper understanding of the Cosmos.

References

Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G. S., Davis, A., Dean, J., Devin, M., Ghemawat, S., Goodfellow, I., Harp, A., Irving, G., Isard, M., Jia, Y., Jozefowicz, R., Kaiser, L., Kudlur, M., Levenberg, J., Mané, D., Monga, R., Moore, S., Murray, D., Olah, C., Schuster, M., Shlens, J., Steiner, B., Sutskever, I., Talwar, K., Tucker, P., Vanhoucke, V., Vasudevan, V., Viégas, F., Vinyals, O., Warden, P., Wattenberg, M., Wicke, M., Yu, Y. & Zheng, X. (2015), ‘TensorFlow: Large-scale machine learning on heterogeneous systems’. Software available from tensorflow.org.

URL: <https://www.tensorflow.org/>

Akiba, T., Sano, S., Yanase, T., Ohta, T. & Koyama, M. (2019), ‘Optuna: A next-generation hyperparameter optimization framework’.

URL: <https://arxiv.org/abs/1907.10902>

Asgari, M., Mead, A. J. & Heymans, C. (2023), ‘The halo model for cosmology: a pedagogical review’, *The Open Journal of Astrophysics* **6**.

URL: <http://dx.doi.org/10.21105/astro.2303.08752>

Bamba, K., Capozziello, S., Nojiri, S. & Odintsov, S. D. (2012), ‘Dark energy cosmology: the equivalent description via different theoretical models and cosmography tests’, *Astrophysics and Space Science* **342**(1), 155–228.

URL: <http://dx.doi.org/10.1007/s10509-012-1181-8>

Basheer, I. & Hajmeer, M. (2001), ‘Artificial neural networks: Fundamentals, com-

References

- puting, design, and application’, *Journal of microbiological methods* **43**, 3–31.
- Bergstra, J. & Bengio, Y. (2012), ‘Random search for hyper-parameter optimization’, *Journal of Machine Learning Research* **13**(10), 281–305.
URL: <http://jmlr.org/papers/v13/bergstra12a.html>
- Chaurasiya, N., More, S., Ishikawa, S., Masaki, S., Kashino, D. & Okumura, T. (2023), ‘Galaxy-dark matter connection of photometric galaxies from the hsc-ssp survey: Galaxy-galaxy lensing and the halo model’.
URL: <https://arxiv.org/abs/2307.03915>
- Coupon, J., Arnouts, S., van Waerbeke, L., Moutard, T., Ilbert, O., van Uitert, E., Erben, T., Garilli, B., Guzzo, L., Heymans, C., Hildebrandt, H., Hoekstra, H., Kilbinger, M., Kitching, T., Mellier, Y., Miller, L., Scodeggio, M., Bonnett, C., Branchini, E., Davidzon, I., De Lucia, G., Fritz, A., Fu, L., Hudelot, P., Hudson, M. J., Kuijken, K., Leauthaud, A., Le Fèvre, O., McCracken, H. J., Moscardini, L., Rowe, B. T. P., Schrabback, T., Semboloni, E. & Velander, M. (2015), ‘The galaxy–halo connection from a joint lensing, clustering and abundance analysis in the cfhtlens/vipers field’, *Monthly Notices of the Royal Astronomical Society* **449**(2), 1352–1379.
URL: <http://dx.doi.org/10.1093/mnras/stv276>
- Del Popolo, A. & Le Delliou, M. (2017), ‘Small scale problems of the cdm model: A short review’, *Galaxies* **5**(1), 17.
URL: <http://dx.doi.org/10.3390/galaxies5010017>
- Duffy, A. R., Schaye, J., Kay, S. T. & Dalla Vecchia, C. (2008), ‘Dark matter halo concentrations in the wilkinson microwave anisotropy probe year 5 cosmology’, *Monthly Notices of the Royal Astronomical Society: Letters* **390**(1), L64–L68.
URL: <http://dx.doi.org/10.1111/j.1745-3933.2008.00537.x>
- Eisenstein, D. J. & Hu, W. (1998), ‘Baryonic Features in the Matter Transfer Function’, *496*(2), 605–614.

References

Euclid Collaboration, Mellier, Y., Abdurro'uf, Acevedo Barroso, J. A., Achúcarro, A., Adamek, J., Adam, R., Addison, G. E., Aghanim, N., Agüena, M., Ajani, V., Akrami, Y., Al-Bahllawan, A., Alavi, A., Albuquerque, I. S., Alestas, G., Alguero, G., Allaoui, A., Allen, S. W., Allevalo, V., Alonso-Tetilla, A. V., Altieri, B., Alvarez-Candal, A., Alvi, S., Amara, A., Amendola, L., Amiaux, J., Andika, I. T., Andreon, S., Andrews, A., Angora, G., Angulo, R. E., Annibali, F., Anselmi, A., Anselmi, S., Arcari, S., Archidiacono, M., Aricò, G., Arnaud, M., Arnouts, S., Asgari, M., Asorey, J., Atayde, L., Atek, H., Atrio-Barandela, F., Aubert, M., Aubourg, E., Auphan, T., Auricchio, N., Aussel, B., Aussel, H., Avelino, P. P., Avgoustidis, A., Avila, S., Awan, S., Azzollini, R., Baccigalupi, C., Bachelet, E., Bacon, D., Baes, M., Bagley, M. B., Bahr-Kalus, B., Balaguera-Antolinez, A., Balbinot, E., Balcells, M., Baldi, M., Baldry, I., Balestra, A., Ballardini, M., Ballester, O., Balogh, M., Bañados, E., Barbier, R., Bardelli, S., Baron, M., Barreiro, T., Barrena, R., Barriere, J. C., Barros, B. J., Barthelémy, A., Bartolo, N., Basset, A., Battaglia, P., Battisti, A. J., Baugh, C. M., Baumont, L., Bazzanini, L., Beaulieu, J. P., Beckmann, V., Belikov, A. N., Bel, J., Bellagamba, F., Bella, M., Bellini, E., Benabed, K., Bender, R., Benevento, G., Bennett, C. L., Benson, K., Bergamini, P., Bermejo-Climent, J. R., Bernardeau, F., Bertacca, D., Berthe, M., Berthier, J., Bethermin, M., Beutler, F., Bevillon, C., Bhargava, S., Bhatawdekar, R., Bianchi, D., Bisigello, L., Biviano, A., Blake, R. P., Blanchard, A., Blazek, J., Blot, L., Bosco, A., Bodendorf, C., Boenke, T., Böhringer, H., Boldrini, P., Bolzonella, M., Bonchi, A., Bonici, M., Bonino, D., Bonino, L., Bonvin, C., Bon, W., Booth, J. T., Borgani, S., Borlaff, A. S., Borsato, E., Bose, B., Botticella, M. T., Boucaud, A., Bouche, F., Boucher, J. S., Boutigny, D., Bouvard, T., Bouwens, R., Bouy, H., Bowler, R. A. A., Bozza, V., Bozzo, E., Branchini, E., Brando, G., Brau-Nogue, S., Brekke, P., Bremer, M. N., Brescia, M., Breton, M. A., Brinchmann, J., Brinckmann, T., Brockley-Blatt, C., Brodwin, M., Brouard, L., Brown, M. L., Bruton, S., Bucko, J., Buddelmeijer, H., Buenadicha, G., Buitrago, F., Burger, P., Burigana, C., Busillo, V., Busonero,

References

- D., Cabanac, R., Cabayol-Garcia, L., Cagliari, M. S., Caillat, A., Caillat, L., Calabrese, M., Calabro, A., Calderone, G., Calura, F., Camacho Quevedo, B., Camera, S., Campos, L., Cañas-Herrera, G., Candini, G. P., Cantiello, M., Capobianco, V., Cappellaro, E., Cappelluti, N., Cappi, A., Caputi, K. I., Cara, C., Carbone, C., Cardone, V. F., Carella, E., Carlberg, R. G., Carle, M., Carminati, L., Caro, F., Carrasco, J. M., Carretero, J., Carrilho, P., Carron Duque, J. & Carry, B. (2025), ‘Euclid: I. Overview of the Euclid mission’, **697**, A1.
- Field, S. E., Galley, C. R., Hesthaven, J. S., Kaye, J. & Tiglio, M. (2014), ‘Fast prediction and evaluation of gravitational waveforms using surrogate models’, *Physical Review X* **4**(3).
URL: <http://dx.doi.org/10.1103/PhysRevX.4.031006>
- Goodfellow, I., Bengio, Y. & Courville, A. (2016), *Deep Learning*, MIT Press.
URL: <http://www.deeplearningbook.org>
- Kasim, M. F., Watson-Parris, D., Deaconu, L., Oliver, S., Hatfield, P., Froula, D. H., Gregori, G., Jarvis, M., Khatiwala, S., Korenaga, J., Topp-Muggleston, J., Viezzer, E. & Vinko, S. M. (2021), ‘Building high accuracy emulators for scientific simulations with deep neural architecture search’, *Machine Learning: Science and Technology* **3**(1), 015013.
URL: <http://dx.doi.org/10.1088/2632-2153/ac3ffa>
- Kingma, D. P. & Ba, J. (2017), ‘Adam: A method for stochastic optimization’. Presented at the 3rd International Conference on Learning Representations (ICLR 2015).
URL: <https://arxiv.org/abs/1412.6980>
- Krasnopolsky, V., Nadiga, S., Mehra, A., Bayler, E. & Behringer, D. (2015), ‘Neural networks technique for filling gaps in satellite measurements: Application to ocean color observations’, *Computational Intelligence and Neuroscience* **2016**, 1–9.
- Lewis, A. & Bridle, S. (2002), ‘Cosmological parameters from cmb and other data:

References

- A monte carlo approach', *Physical Review D* **66**(10).
URL: <http://dx.doi.org/10.1103/PhysRevD.66.103511>
- Linke, L., Simon, P., Schneider, P., Farrow, D. J., Rödiger, J. & Wright, A. H. (2022), 'Kids+viking+gama: Halo occupation distributions and correlations of satellite numbers with a new halo model of the galaxy-matter bispectrum for galaxy-galaxy-galaxy lensing', *Astronomy & Astrophysics* **665**, A38.
URL: <http://dx.doi.org/10.1051/0004-6361/202243711>
- Mo, H. J. & White, S. D. M. (1994), 'An analytic model for the gravitational clustering of dark matter haloes'.
URL: <https://arxiv.org/abs/astro-ph/9412088>
- Navarro, J. F., Frenk, C. S. & White, S. D. M. (1997), 'A Universal Density Profile from Hierarchical Clustering', **490**(2), 493–508.
- Navrátil, J., King, A., Rios, J., Kollias, G., Torrado, R. & Codas, A. (2019), 'Accelerating physics-based simulations using end-to-end neural network proxies: An application in oil reservoir modeling', *Frontiers in Big Data* **2**.
URL: <http://dx.doi.org/10.3389/fdata.2019.00033>
- Persic, M., Salucci, P. & Stel, F. (1996), 'The universal rotation curve of spiral galaxies — I. The dark matter connection', **281**(1), 27–47.
- Piras, D., Polanska, A., Mancini, A. S., Price, M. A. & McEwen, J. D. (2024), 'The future of cosmological likelihood-based inference: accelerated high-dimensional parameter estimation and model comparison', *The Open Journal of Astrophysics* .
URL: <http://dx.doi.org/10.33232/001c.123368>
- Planck Collaboration: Aghanim, N., Akrami, Y., Arroja, F., Ashdown, M., Aumont, J., Baccigalupi, C., Ballardini, M., Banday, A. J., Barreiro, R. B., Bartolo, N., Basak, S., Battye, R., Benabed, K., Bernard, J.-P., Bersanelli, M., Bielewicz, P., Bock, J. J., Bond, J. R., Borrill, J., Bouchet, F. R., Boulanger, F., Bucher, M.,

References

Burigana, C., Butler, R. C., Calabrese, E., Cardoso, J.-F., Carron, J., Casaponsa, B., Challinor, A., Chiang, H. C., Colombo, L. P. L., Combet, C., Contreras, D., Crill, B. P., Cuttaia, F., de Bernardis, P., de Zotti, G., Delabrouille, J., Delouis, J.-M., Désert, F.-X., Di Valentino, E., Dickinson, C., Diego, J. M., Donzelli, S., Doré, O., Douspis, M., Ducout, A., Dupac, X., Efstathiou, G., Elsner, F., Enßlin, T. A., Eriksen, H. K., Falgarone, E., Fantaye, Y., Fergusson, J., Fernandez-Cobos, R., Finelli, F., Forastieri, F., Frailis, M., Franceschi, E., Frolov, A., Galeotta, S., Galli, S., Ganga, K., Génova-Santos, R. T., Gerbino, M., Ghosh, T., González-Nuevo, J., Górski, K. M., Gratton, S., Gruppuso, A., Gudmundsson, J. E., Hamann, J., Handley, W., Hansen, F. K., Helou, G., Herranz, D., Hildebrandt, S. R., Hivon, E., Huang, Z., Jaffe, A. H., Jones, W. C., Karakci, A., Keihänen, E., Keskitalo, R., Kiiveri, K., Kim, J., Kisner, T. S., Knox, L., Krachmalnicoff, N., Kunz, M., Kurki-Suonio, H., Lagache, G., Lamarre, J.-M., Langer, M., Lasenby, A., Lattanzi, M., Lawrence, C. R., Le Jeune, M., Leahy, J. P., Lesgourgues, J., Levrier, F., Lewis, A., Liguori, M., Lilje, P. B., Lilley, M., Lindholm, V., López-Caniego, M., Lubin, P. M., Ma, Y.-Z., Macías-Pérez, J. F., Maggio, G., Maino, D., Mandolesi, N., Mangilli, A., Marcos-Caballero, A., Maris, M., Martin, P. G., Martinelli, M., Martínez-González, E., Matarrese, S., Mauri, N., McEwen, J. D., Meerburg, P. D., Meinhold, P. R., Melchiorri, A., Mennella, A., Migliaccio, M., Millea, M., Mitra, S., Miville-Deschênes, M.-A., Molinari, D., Moneti, A., Montier, L., Morgante, G., Moss, A., Mottet, S., Münchmeyer, M., Natoli, P., Nørgaard-Nielsen, H. U., Oxborrow, C. A., Pagano, L., Paoletti, D., Partridge, B., Patanchon, G., Pearson, T. J., Peel, M., Peiris, H. V., Perrotta, F., Pettorino, V., Piacentini, F., Polastri, L., Polenta, G., Puget, J.-L., Rachen, J. P., Reinecke, M., Remazeilles, M., Renault, C., Renzi, A., Rocha, G., Rosset, C., Roudier, G., Rubiño-Martín, J. A., Ruiz-Granados, B., Salvati, L., Sandri, M., Savelainen, M., Scott, D., Shellard, E. P. S., Shiraishi, M., Sirignano, C., Sirri, G., Spencer, L. D., Sunyaev, R., Suur-Uski, A.-S., Tauber, J. A., Tavagnacco, D., Tenti, M., Terenzi, L., Toffolatti, L., Tomasi, M., Trombetti, T., Valiviita, J., Van Tent, B., Vibert,

References

- L., Vielva, P., Villa, F., Vittorio, N., Wandelt, B. D., Wehus, I. K., White, M., White, S. D. M., Zacchei, A. & Zonca, A. (2020), ‘Planck2018 results: I. overview and the cosmological legacy of planck’, *Astronomy & Astrophysics* **641**, A1.
URL: <http://dx.doi.org/10.1051/0004-6361/201833880>
- Rumelhart, D. E., Hinton, G. E. & Williams, R. J. (1986), ‘Learning representations by back-propagating errors’, **323**(6088), 533–536.
- Schmidhuber, J. (2015), ‘Deep learning in neural networks: An overview’, *Neural Networks* **61**, 85–117.
URL: <http://dx.doi.org/10.1016/j.neunet.2014.09.003>
- Scoccimarro, R., Sheth, R. K., Hui, L. & Jain, B. (2001), ‘How many galaxies fit in a halo? constraints on galaxy formation efficiency from spatial clustering’, *The Astrophysical Journal* **546**(1), 20–34.
URL: <http://dx.doi.org/10.1086/318261>
- Sheth, R. K. & Tormen, G. (1999), ‘Large-scale bias and the peak background split’, **308**(1), 119–126.
- Snoek, J., Larochelle, H. & Adams, R. P. (2012), Practical bayesian optimization of machine learning algorithms, in ‘Advances in Neural Information Processing Systems 25 (NeurIPS 2012)’.
URL: <https://arxiv.org/abs/1206.2944>
- Spurio Mancini, A., Piras, D., Alsing, J., Joachimi, B. & Hobson, M. P. (2022), ‘Cosmopower: emulating cosmological power spectra for accelerated bayesian inference from next-generation surveys’, *Monthly Notices of the Royal Astronomical Society* **511**(2), 1771–1788.
URL: <http://dx.doi.org/10.1093/mnras/stac064>
- Stock, J., Petersen, T., Behrendt, C.-A., Federrath, H. & Schwaneberg, T. (2022), ‘Privatsphärefreundliches maschinelles lernen: Teil 1: Grundlagen und verfahren’, *Informatik Spektrum* **45**.

References

- Toledo-Marín, J. Q., Fox, G., Sluka, J. P. & Glazier, J. A. (2021), ‘Deep learning approaches to surrogates for solving the diffusion equation for mechanistic real-world simulations’, *arXiv preprint arXiv:2102.05527*.
- URL:** <https://arxiv.org/abs/2102.05527>
- Tripathy, R. K. & Bilonis, I. (2018), ‘Deep UQ: Learning deep neural network surrogate models for high dimensional uncertainty quantification’, *Journal of Computational Physics* **375**, 565–588.
- URL:** <http://dx.doi.org/10.1016/j.jcp.2018.08.036>
- Vegetti, S., Birrer, S., Despali, G., Fassnacht, C. D., Gilman, D., Hezaveh, Y., Levasseur, L. P., McKean, J. P., Powell, D. M., O’Riordan, C. M. & Vernardos, G. (2023), ‘Strong gravitational lensing as a probe of dark matter’.
- URL:** <https://arxiv.org/abs/2306.11781>
- Zacharegkas, G., Chang, C., Prat, J., Hartley, W., Mucesh, S., Alarcon, A., Alves, O., Amon, A., Bechtol, K., Becker, M. R., Bernstein, G., Blazek, J., Campos, A., Rosell, A. C., Kind, M. C., Cawthon, R., Chen, R., Choi, A., Cordero, J., Davis, C., Derose, J., Diehl, H., Dodelson, S., Doux, C., Drlica-Wagner, A., Eckert, K., Eifler, T. F., Elvin-Poole, J., Everett, S., Fang, X., Ferte, A., Gatti, M., Giannini, G., Gruen, D., Gruendl, R. A., Harrison, I., Huang, H., Huff, E. M., Jarvis, M., Krause, E., Kuropatkin, N., Leget, P. F., Maccrann, N., McCullough, J., Myles, J., Alsina, A. N., Pandey, S., Raveri, M., Rollins, R. P., Roodman, A., Ross, A. J., Rykoff, E. S., Sanchez, C., Secco, L. F., Sevilla-Noarbe, I., Sheldon, E., Shin, T., Troxel, M. A., Tutusaus, I., Yanny, B., Yin, B., Zhang, Y. & Zuntz, J. (2025), ‘Constraining the stellar-to-halo mass relation with galaxy clustering and weak lensing from des year 3 data’.
- URL:** <https://arxiv.org/abs/2506.22367>
- Zehavi, I., Zheng, Z., Weinberg, D. H., Frieman, J. A., Berlind, A. A., Blanton, M. R., Scoccimarro, R., Sheth, R. K., Strauss, M. A., Kayo, I., Suto, Y., Fukugita, M., Nakamura, O., Bahcall, N. A., Brinkmann, J., Gunn, J. E., Hennessey, G. S.,

References

Ivezić, , Knapp, G. R., Loveday, J., Meiksin, A., Schlegel, D. J., Schneider, D. P., Szapudi, I., Tegmark, M., Vogeley, M. S. & York, D. G. (2005), ‘The luminosity and color dependence of the galaxy correlation function’, *The Astrophysical Journal* **630**(1), 1–27.

URL: <http://dx.doi.org/10.1086/431891>

Zhang, Y., Tino, P., Leonardis, A. & Tang, K. (2021), ‘A survey on neural network interpretability’, *IEEE Transactions on Emerging Topics in Computational Intelligence* **5**(5), 726–742.

URL: <http://dx.doi.org/10.1109/TETCI.2021.3100641>

Zheng, Z., Coil, A. L. & Zehavi, I. (2007), ‘Galaxy evolution from halo occupation distribution modeling of deep2 and sdss galaxy clustering’, *The Astrophysical Journal* **667**(2), 760–779.

URL: <http://dx.doi.org/10.1086/521074>